

Final Exam Review

DISCLAIMER !!!!!

Contained herein are some of the important concepts that many of you struggled with on the homework assignments that will be important to understand for the exam

This review **does not** contain everything you need to review !!

(Review homework, study proton, take practice test, consult study guide, watch habits review videos, skim tutorials.)

Difficult Probability Concepts To Review

Vocab:

Mutually Exclusive:

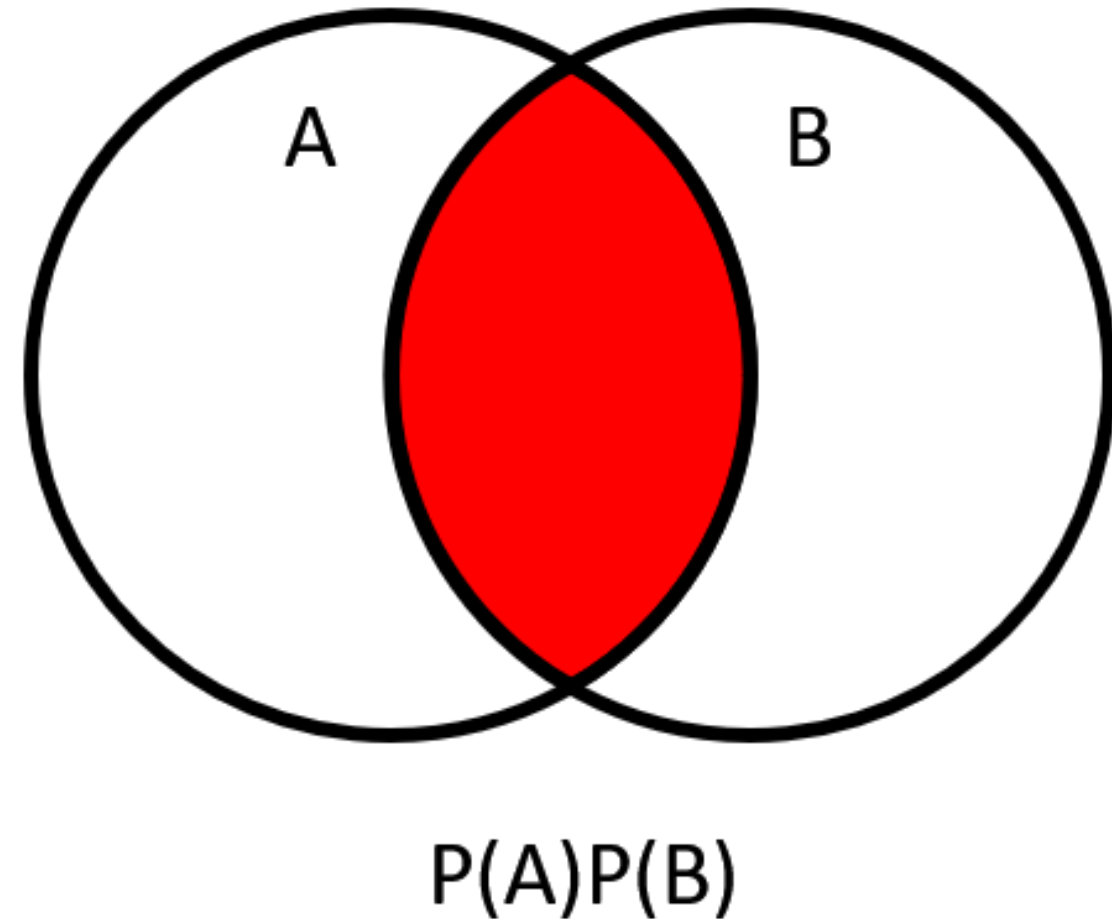
Two or more events that can't happen at the same time or on the same trial.

- P(events A & B Occurring)
 - If A and B are independent
 - $P(A) * P(B)$
- P(event A OR B Occurring)
 - If A and B **are** mutually exclusive
 - $P(A \text{ or } B) = p(A) + p(B)$
 - If A and B are **not** mutually exclusive
 - $P(A \text{ or } B) = p(A) + p(B) - p(A \text{ and } B)$
- P(event A happening at least once)
 - $P(A \text{ happening at least once}) = 1 - (1 - P(A \text{ will happen}))$
 - $P(A \text{ happening at least once in 15 tries}) = 1 - ((1 - (P(A \text{ will happen})))^{15})$

These are the types of probability calculations most of you had trouble solving on the pop quiz.

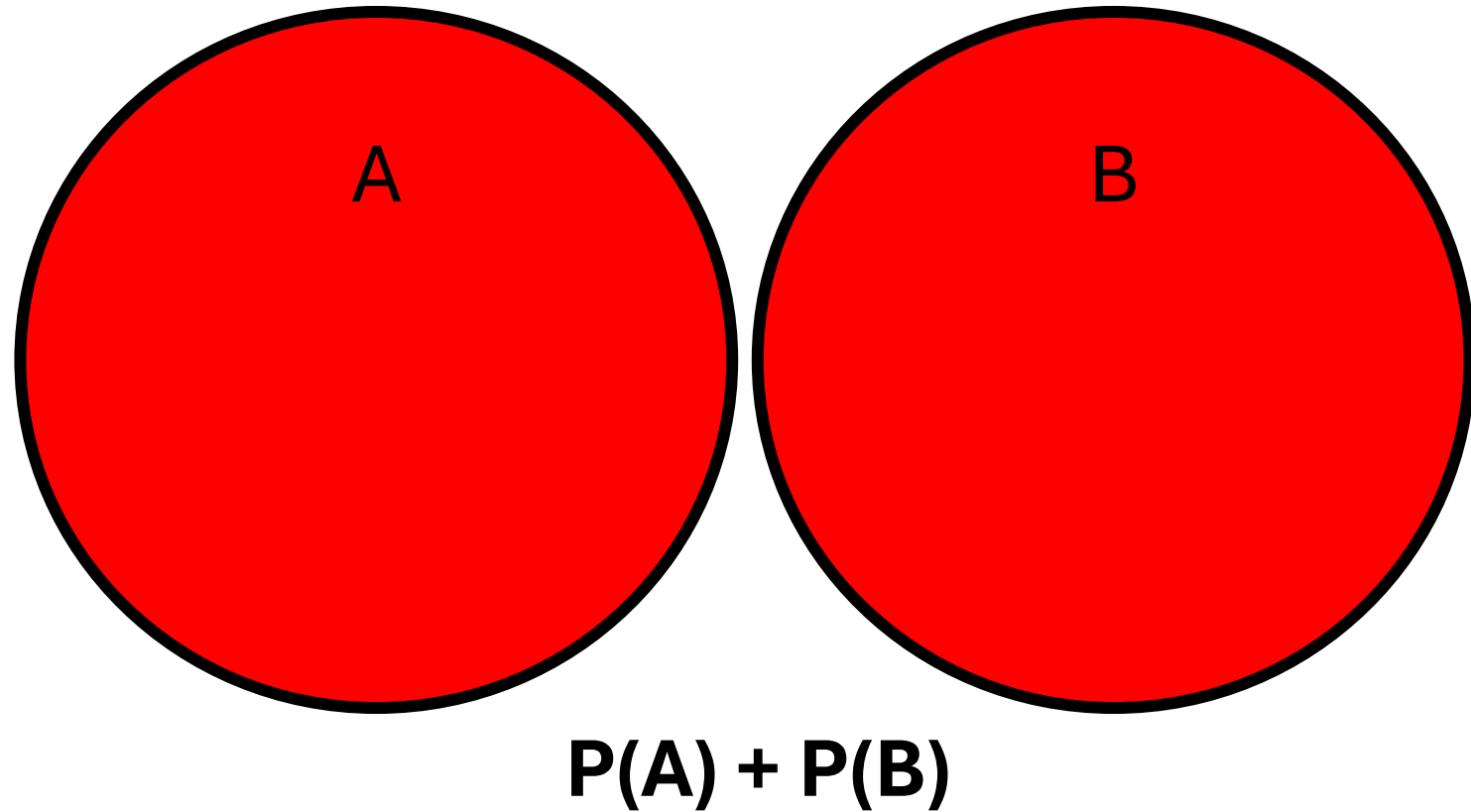
P (A & B)

- Probability of two events happening at the same time.
- If the two events are **independent** (A happening does not affect the chance of B happening and vice versa)
 - Calculate the probability of both occurring by multiplying the probability of A and B
 - $P(A \& B) = P(A) * P(B)$



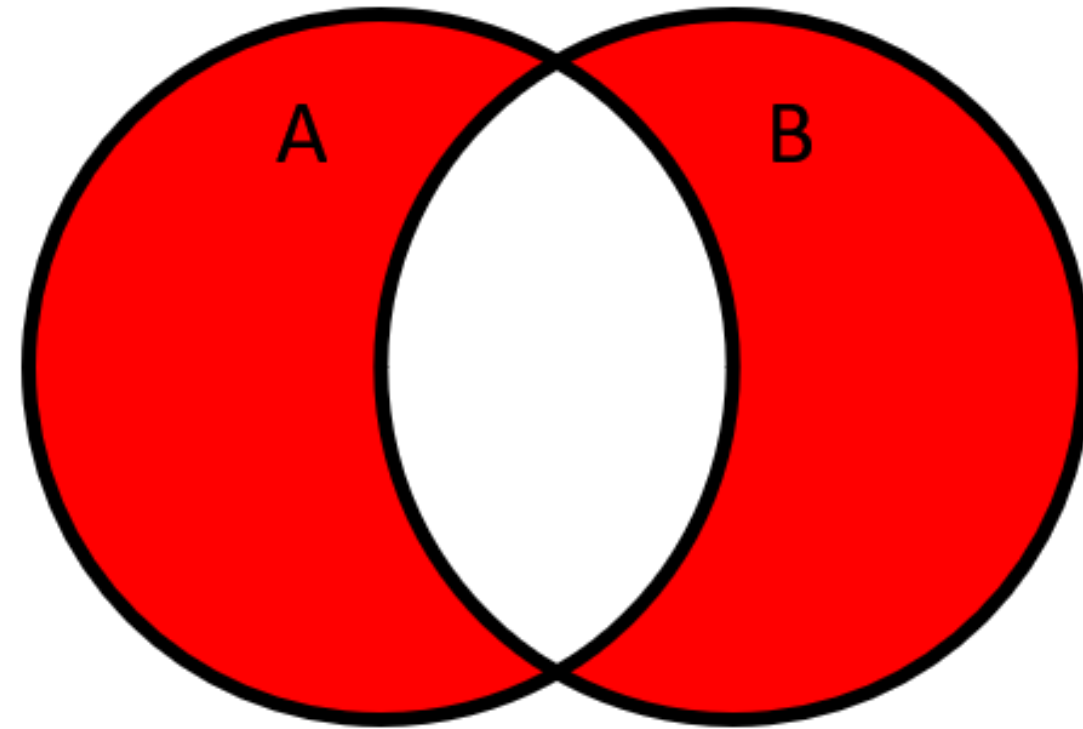
$P(A \text{ OR } B)$ (Mutually Exclusive)

- Addition Rule:
 - $P(A \text{ OR } B) = P(A) + P(B)$



$P(A \text{ OR } B)$ (Not Mutually Exclusive)

- Addition Rule
- Probability of event A happening ($P(A)$) OR event B happening ($P(B)$) without double counting the overlap ($- P(A)*P(B)$)
- $P(A \text{ OR } B) = P(A) + P(B) - (P(A)*P(B))$



$$P(A) + P(B) - P(A)P(B)$$

Question 3

(i) Given the weather forecast shown below, calculate the probability that it will rain EVERY day next week. Assume that rain events on any given day are independent of rain events on any other day. **(ii) Given the weather forecast shown below, calculate the probability that it will rain at least one day in the next week.** Assume that rain events on any given day are independent of rain events on any other day.

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
							
Chance of rain	10%	20%	30%	30%	20%	5%	5%

Question 3

(i) Given the weather forecast shown below, calculate the probability that it will rain EVERY day next week. Assume that rain events on any given day are independent of rain events on any other day. **(ii) Given the weather forecast shown below, calculate the probability that it will rain at least one day in the next week.** Assume that rain events on any given day are independent of rain events on any other day.

+

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
							
Chance of rain	10%	20%	30%	30%	20%	5%	5%

□

- (i) $P(\text{rain every day}) = 0.1 \times 0.2 \times 0.3 \times 0.3 \times 0.2 \times 0.05 \times 0.05 = 9 \times 10^{-7}$ - tiny probability, about 1 in a million.
- (ii) $P(\text{rain at least one day}) = 1 - (0.9 \times 0.8 \times 0.7 \times 0.7 \times 0.8 \times 0.95 \times 0.95) = 0.75$ (fairly large probability - 75%)

Homework Review

Molecules and Life & Climate and Us

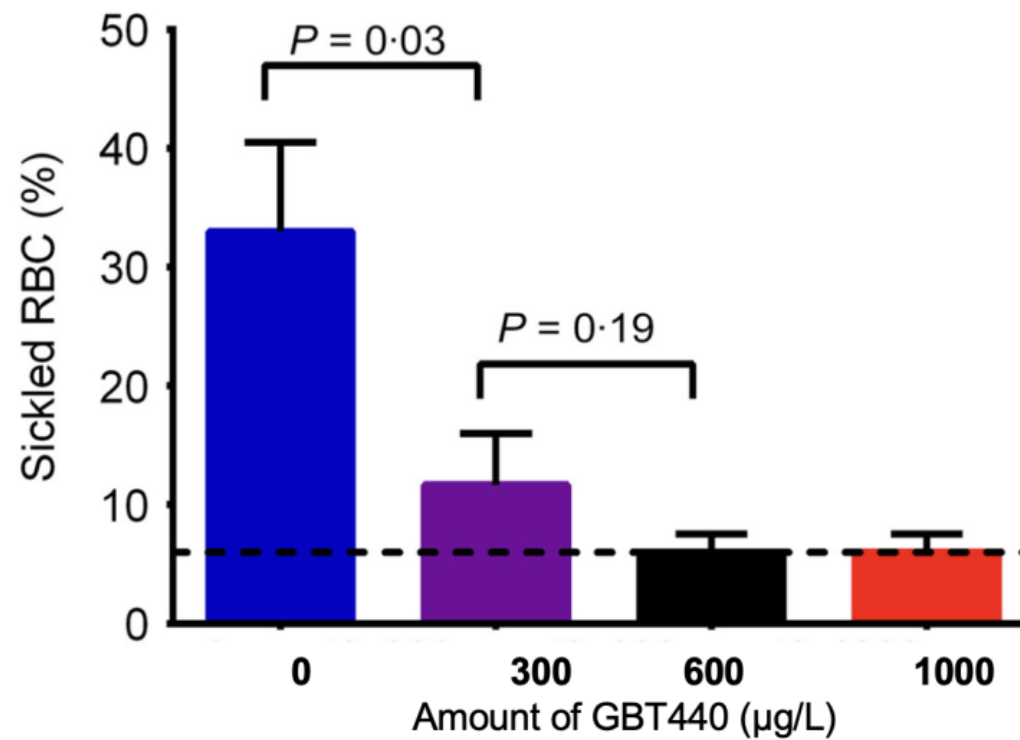


Figure 2. Percentage of red blood cells (RBC) that had a sickle shape after the addition of various amounts of GBT440. The p-values for select comparisons are listed above the corresponding brackets. The dotted line indicates the percentage of irreversibly sickled RBCs.

1E. (ii) Based on the results in Figure 2, does increasing the amount of GBT440 change its effect on the sickling of red blood cells? Explain using your knowledge of statistics.

- No, there is not a significant difference between the percentage of sickled red blood cells with 300 and 600 µg/L of GBT440, shown by the p-value of 0.19 (greater than 0.05). At higher doses, the only sickled red blood cells left are those that are irreversibly sickled.*

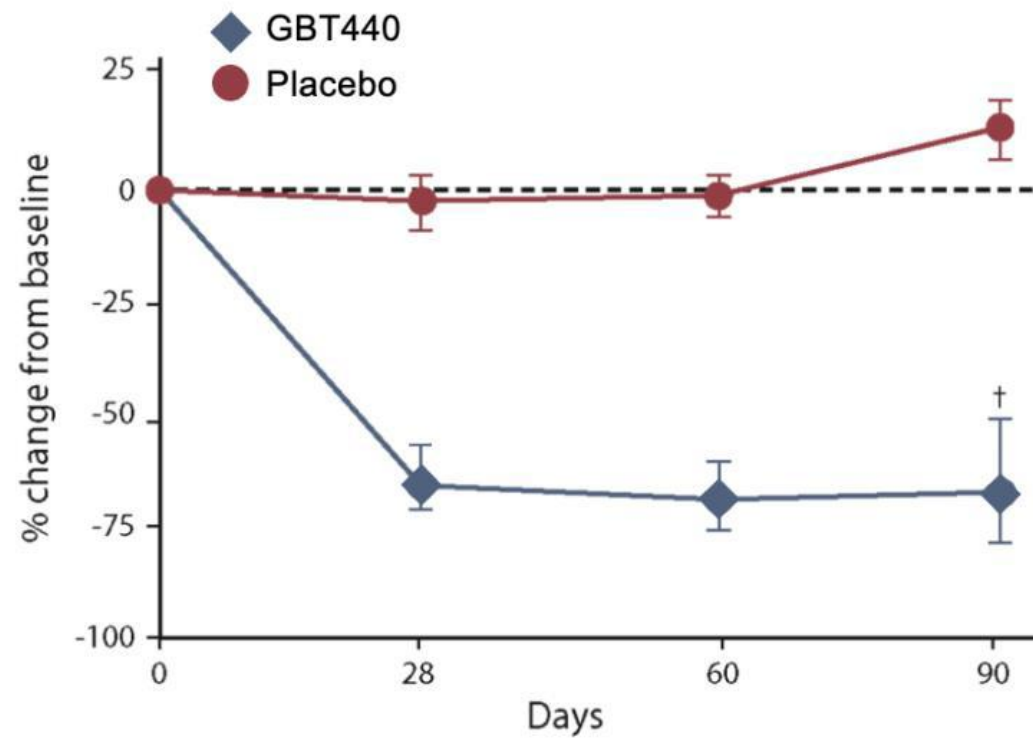


Figure 3. Percentage change from baseline level of sickled red blood cells in patients given GBT440 compared to those given a placebo containing all the same components as the GBT440 treatment except the compound itself. Error bars represent the 95% confidence intervals.

1F (ii). Based on Figure 3, is GBT440 effective at reducing the amount of sickled red blood cells in patients? Explain using your knowledge of statistics.

Yes, the GBT440 treatment significantly reduces the amount of sickled red blood cells compared to baseline and compared to the patients receiving the placebo. The difference is significant because the 95% confidence intervals don't overlap.

		Second Position				
		U	C	A	G	
First Position	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA STOP UAG STOP	UGU } Cys UGC } UGA STOP UGG Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G
						Third Position

Healthy HBB sequence: Leu - Ser - Gly - **Glu** - Pro - Leu
Mutated HBB sequence: Leu - Ser - Gly - **Val** - Pro - Leu

What would the mRNA sequence look like ?

Healthy HBB sequence: CUC – AGU – GGC – **GAA** – CCA - CUA
Mutated HBB sequence: CUC – AGU – GGC – **GUA** – CCA - CUA

1C. (i) Using the mRNA codon chart provided, fill in the blanks in the table below.

	Healthy	Mutated
Affected amino acid	<i>Glu (Glutamate)</i>	<i>Val (valine)</i>
Possible mRNA codons	<i>GAA, GAG</i>	<i>GUU, GUC, GUA, GUG</i>
Complementary DNA codons	<i>CTT, CTC</i>	<i>CAA, CAG, CAT, CAC</i>

Homework Review

2A. (iv) What are the odds that the *HBB* gene in a given cell will experience damage over the course of one year?

This is the probability of a mutation not happening once a day in 365 tries:

$$P(\text{A not happening}) = 1 - 7 \cdot 10^{-3}$$

$$P(\text{A not happening in 365 Tries}) = (1 - 7 \cdot 10^{-3})^{365}$$

$$P(\text{A happening in 365 Tries}): 1 - ((1 - 7 \cdot 10^{-3})^{365}) = 0.923$$

1A. (ii) Which of the following accurately describes Cas9 and its function in genome editing?

- a. A protein responsible for DNA methylation.
- b. A virus that infects bacteria and introduces new genetic material.
- c. A protein enzyme that cuts both strands of DNA at sites specified by an RNA guide.
- d. A small RNA molecule that serves as a template for DNA replication.

C is correct.

1A. (iii) What effect does incorporating a premature STOP codon into the BRCA gene have on the BRCA proteins that are expressed from that gene?

- a) Incorporating the STOP codon deletes the entire BRCA gene, eliminating all BRCA protein production.
- b) Incorporating the STOP codon into the BRCA gene sequence stops translation early and leads to the production of a truncated version of BRCA protein that is non-functional.
- c) Incorporating the STOP codon enhances the production of fully functional BRCA proteins.
- d) Incorporating the STOP codon has no impact on the translation process, as the cellular machinery ignores the premature STOP signal

B is correct.

Homework review

1B. (iv) Estimate the probability that they can successfully edit all four target mutations in every cell in a pool of 50 pancreatic cells.

$$P(\text{all 50 cells edited}) = P(\text{all 4 targets in one cell edited})^{50} = (0.85^4)^{50} = 7.65 \times 10^{-15}$$

1B. (v) What is the probability that they can successfully edit at least one of the four target mutations in every cell in a pool of 50 pancreatic cells? [Hint: $P = 1 - (1 - P_{\text{one}})^n$]

$$\text{Probability of one cell having at least one mutation} = 1 - (1 - 0.85)^4 = 0.9995$$

$$\text{Probability of all 50 cells having at least one mutation} = (0.9995)^{50} = 0.975 \text{ or } 97.5\%$$

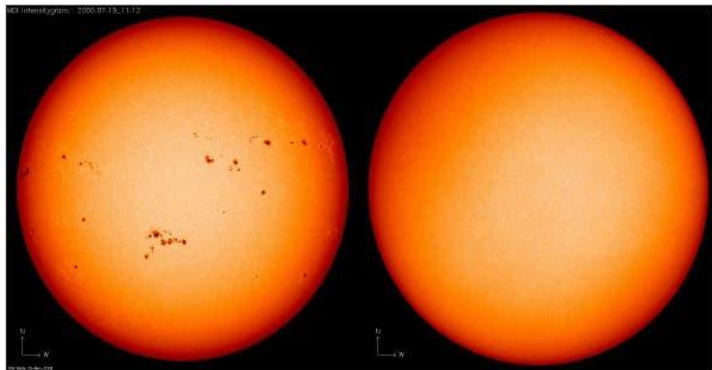
Climate and Us

Lecture Review

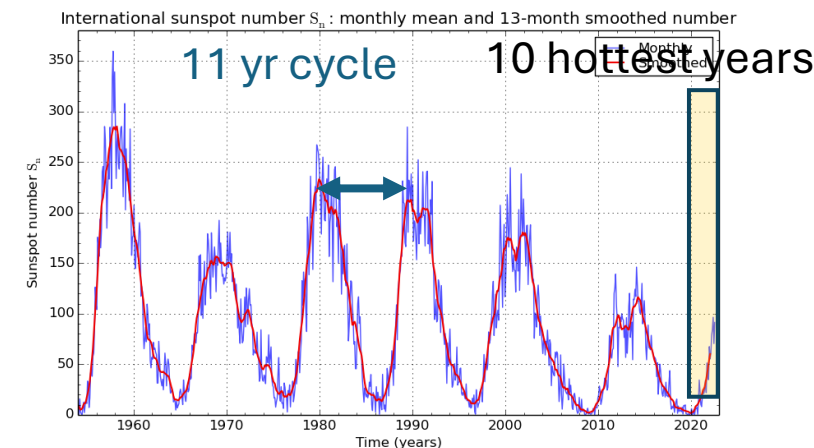
Astronomical Influences - Sunspots

Disturbances in Sun's magnetic field, which can generate energetic solar events

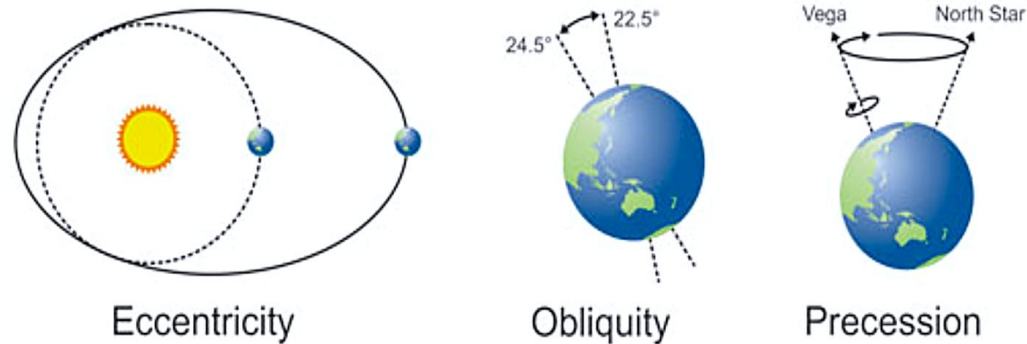
- Yearly avg. number of sunspots increases during periods of high solar activity
- The change in the Sun's yearly average total irradiance during an 11-year cycle is about 0.1 percent or 1.4 W/m^2
- Past decade is *lowest* solar activity recorded this century. Yet, 10 out of 10 hottest years in century during this period



Sunspots near solar maximum on July 19, 2000, and near solar minimum on March 18, 2009 (by NASA)



Astronomical Influences - Orbital Cycles



Systematic changes in the distance and orientation of Earth relative to the Sun
→ affects the amount of solar radiation (insolation) Earth receives

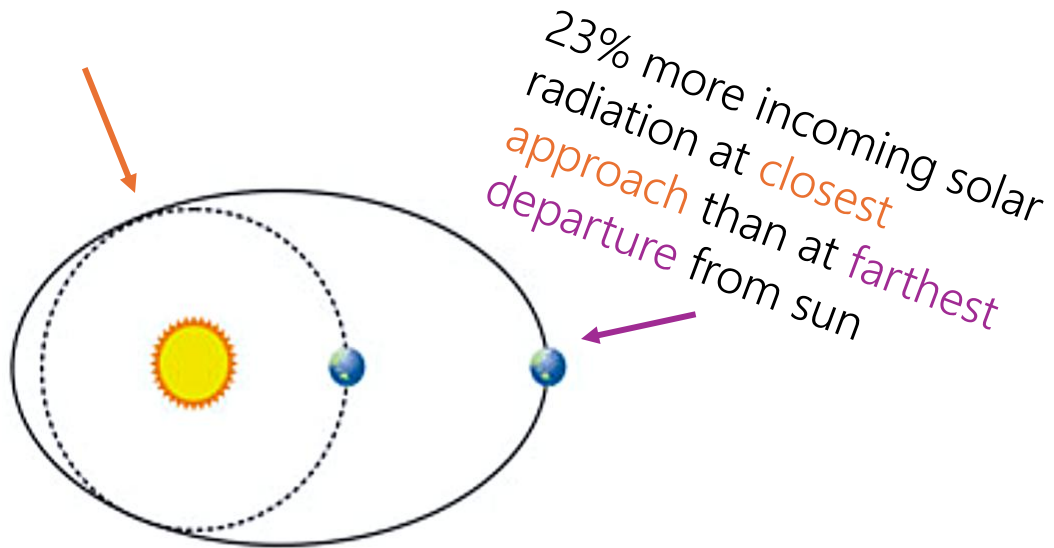
- Affect Earth's climate over 20,000 to 100,000 yrs
- Orbital cycles (aka **Milanković Cycles**):
 - **Eccentricity**: shape of Earth's orbit (100,000 yr cycle)
 - **Obliquity**: angle that Earth's axis is tilted wrt its orbital plane (41,000 yr cycle)
 - **Precession**: direction Earth's axis of rotation is pointed ("wobble") (23,000 yr cycles)

BUT the effect is NOT relevant to the recent global warming we are experiencing

Eccentricity: 100,000 year cycle

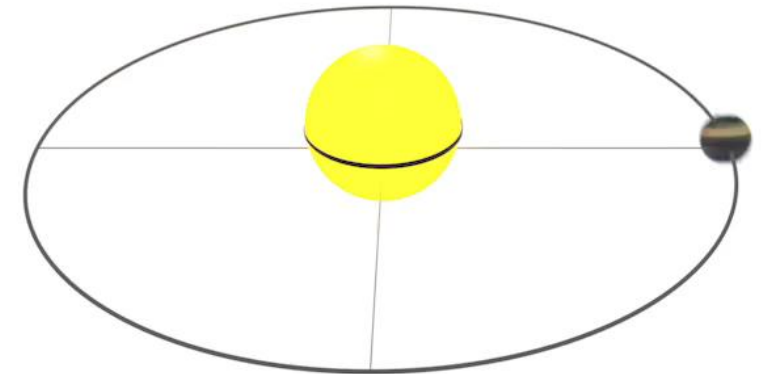
Shape of Earth's orbit from circular to elliptical

- High eccentricity: large seasonal discrepancy in solar radiation
- Currently Earth's eccentricity is nearly circular
 - 5 million km closer to sun on Jan 3rd than on July 4th



Changes in Eccentricity (Orbit Shape)

100,000-year cycles

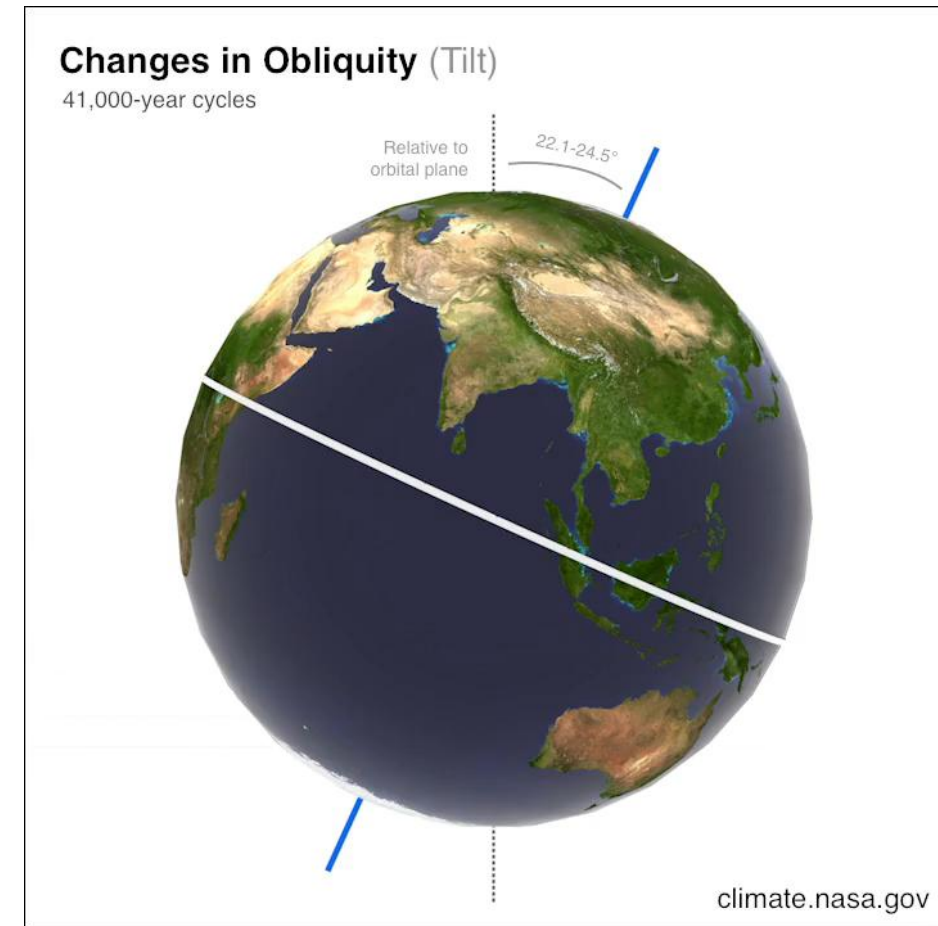


*Changes in eccentricity exaggerated so the effect can be seen. Earth's orbit shape varies between 0.0034 (almost a perfect circle) to 0.058 (slightly elliptical).

Obliquity: 41,000 year cycle

Angle of Earth's axis of rotation is tilted with respect to Earth's orbital plane

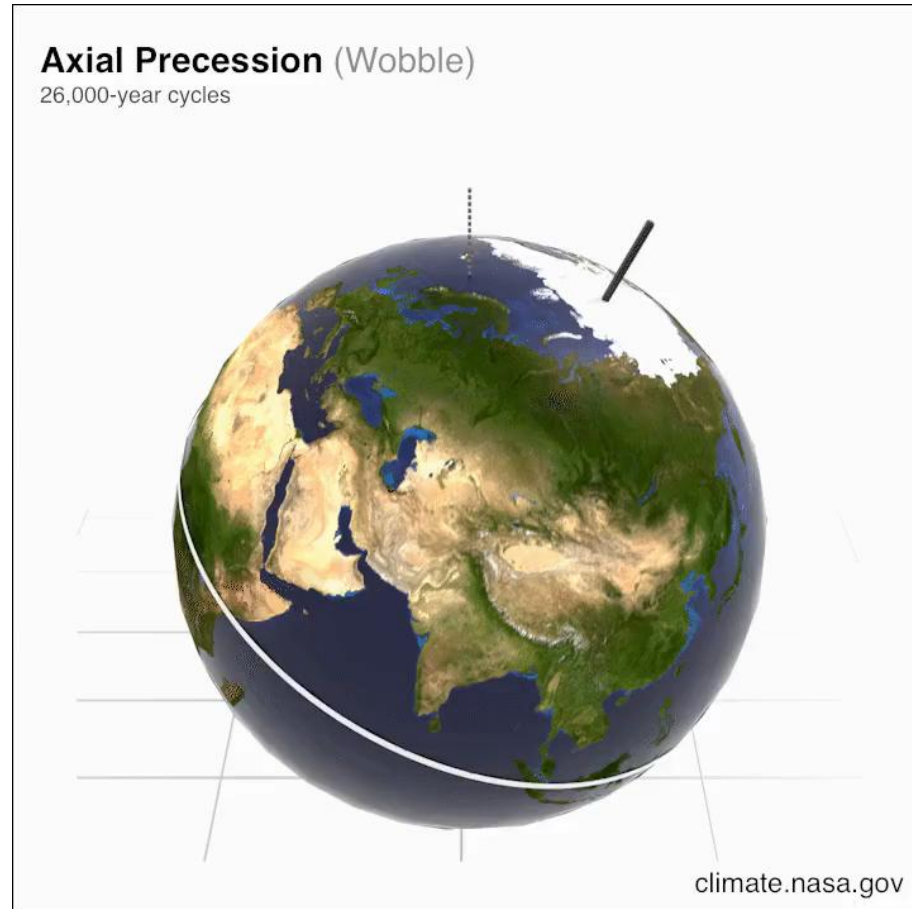
- Varies from 22.1 to 24.5 degrees (currently 23.4 degrees, ~halfway)
- Tilt is the reason for Earth's seasons
 - The higher the tilt angle, the more extreme our seasons are
- Effects aren't uniform globally
- As obliquity decreases, winters will be warmer and summers will be cooler



Precession: 23,000 year cycle

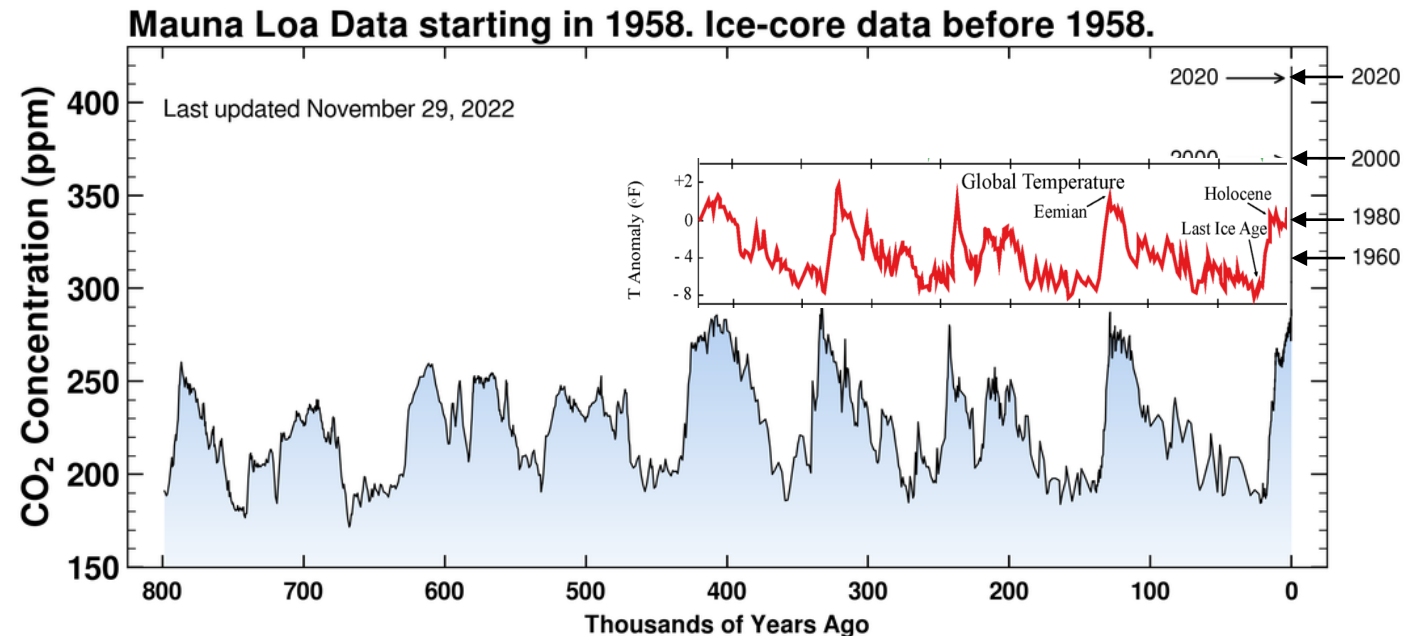
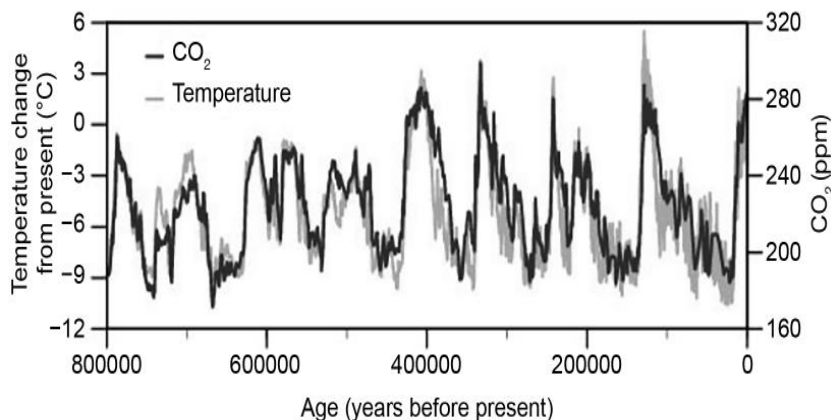
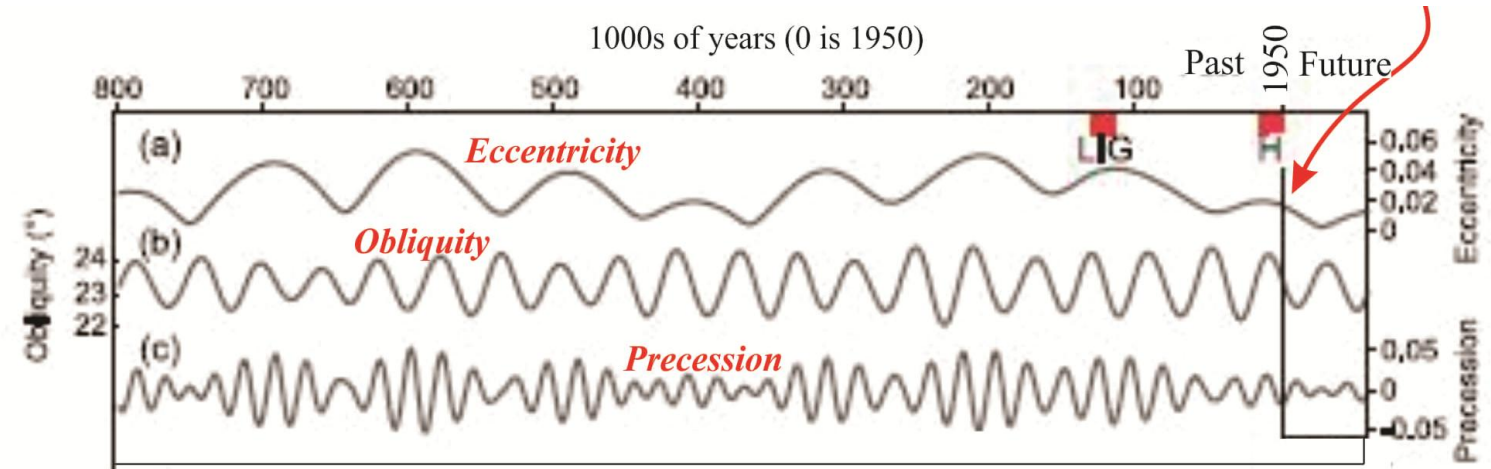
Direction Earth's axis of rotation is pointed

- Seasonal contrasts are more extreme in one hemisphere
 - Currently: S. Hem sees more extremes, while N. hem is more moderate
 - In 13,000 yrs: N. Hem will see more extremes
- Caused by tidal forces
- North stars will eventually change



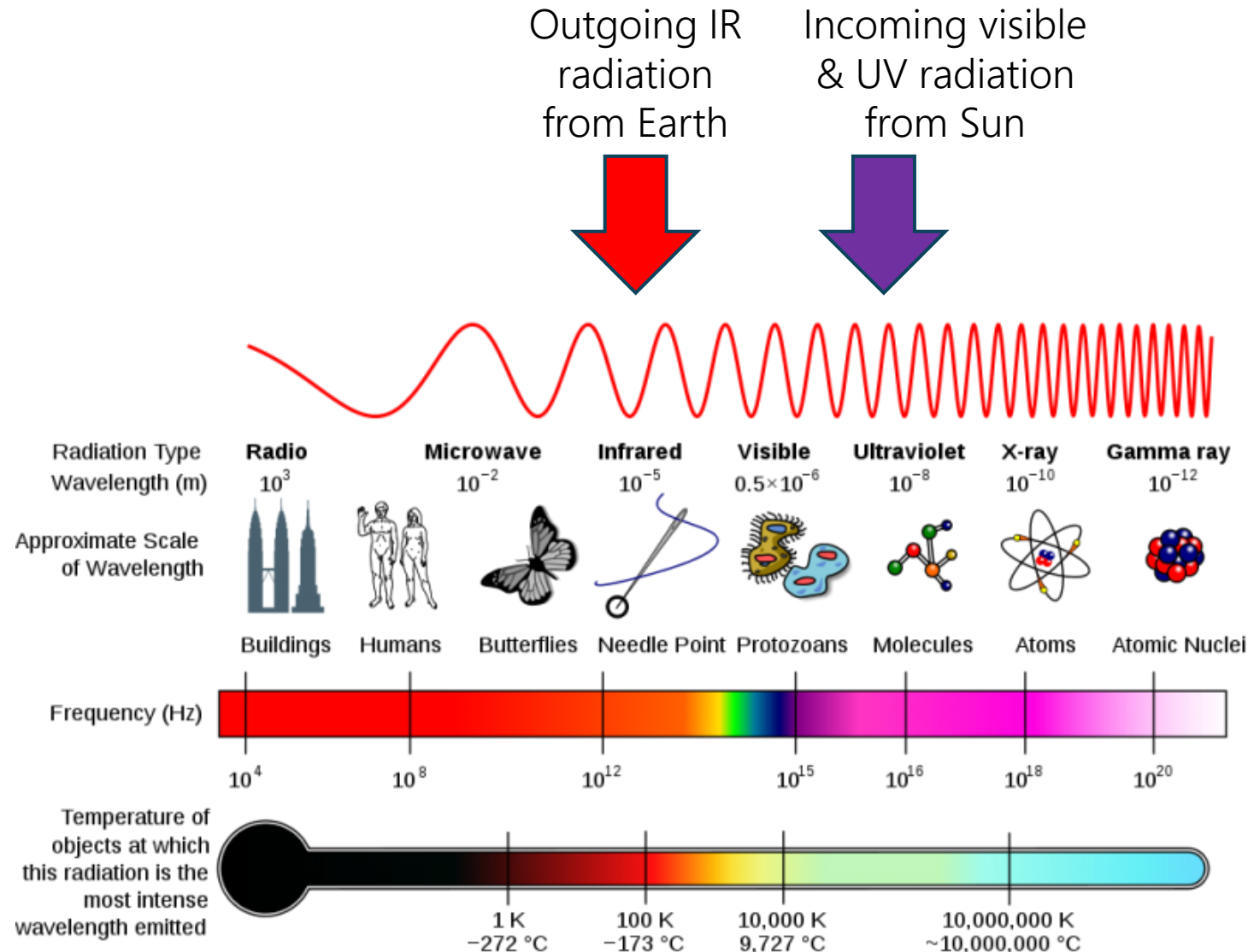
Summary: Orbital Cycles and Climate

- Orbital cycles = periods of glaciation and deglaciation
 - Cycles of T and CO₂
- But long timescales (23-100 kyr)= little difference in last century
- → Cannot explain rapid rise in CO₂ and temp in last century



Quick background: Electromagnetic spectrum

- Electromagnetic energy travels in waves, with different wavelengths
- *Thermal* radiation has different wavelengths depending on temp.
 - Thermal radiation of animals is in the IR spectrum (why we call IR radiation “heat”)
- Solar radiation is **UV** (short wavelength)
- Earth absorbs and then re-emits this energy as “heat” in the **IR spectrum** (long wavelength)



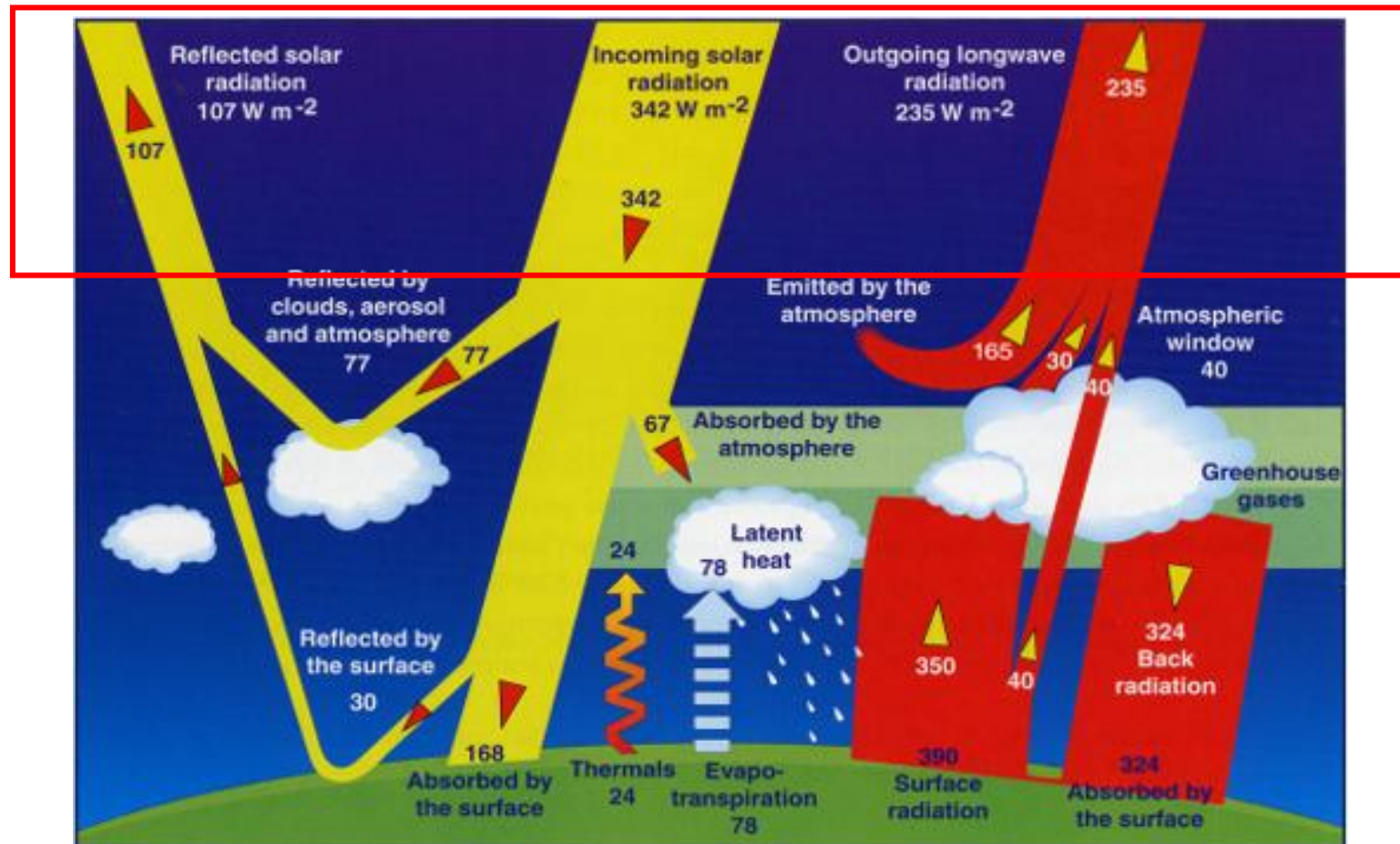
Earth's Energy Balance

Earth's energy balance: The balance between incoming and outgoing solar radiation

Energy In (Shortwave)

=

Energy Out (Longwave)



Earth's Energy Balance

When energy comes in, where does it go?

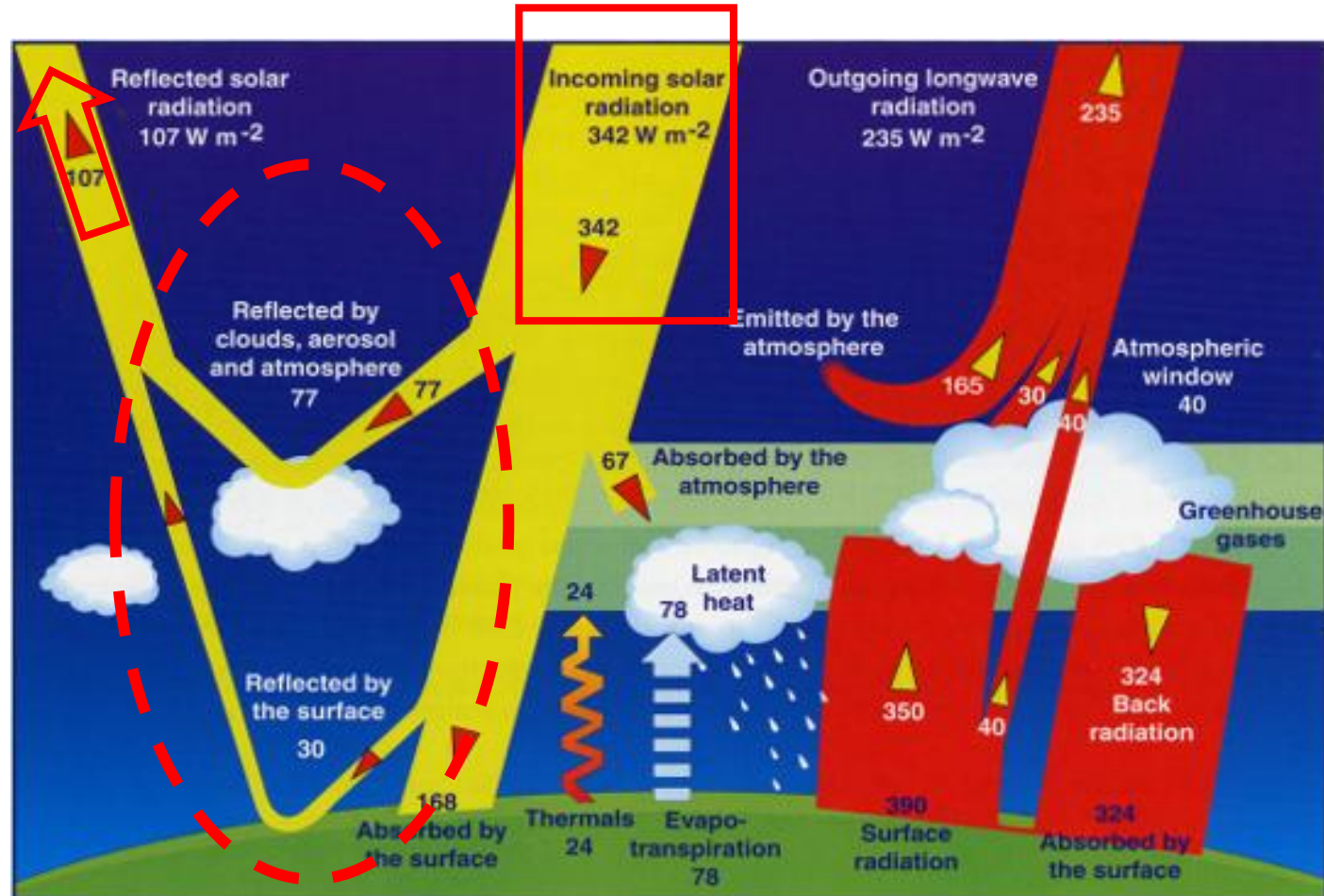
Albedo

Net effect of the **reflectivity** of Earth - makes the Earth **an imperfect absorber**

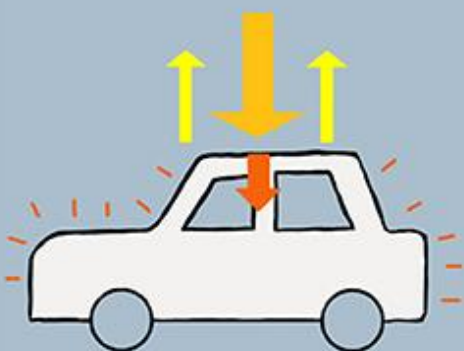
Reflectivity of Earth:
31% of light is reflected back to space

Albedo examples:

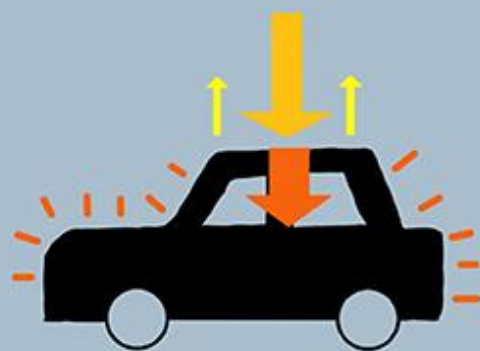
High: ice, snow, clouds
Low: forests, pavements



High Albedo



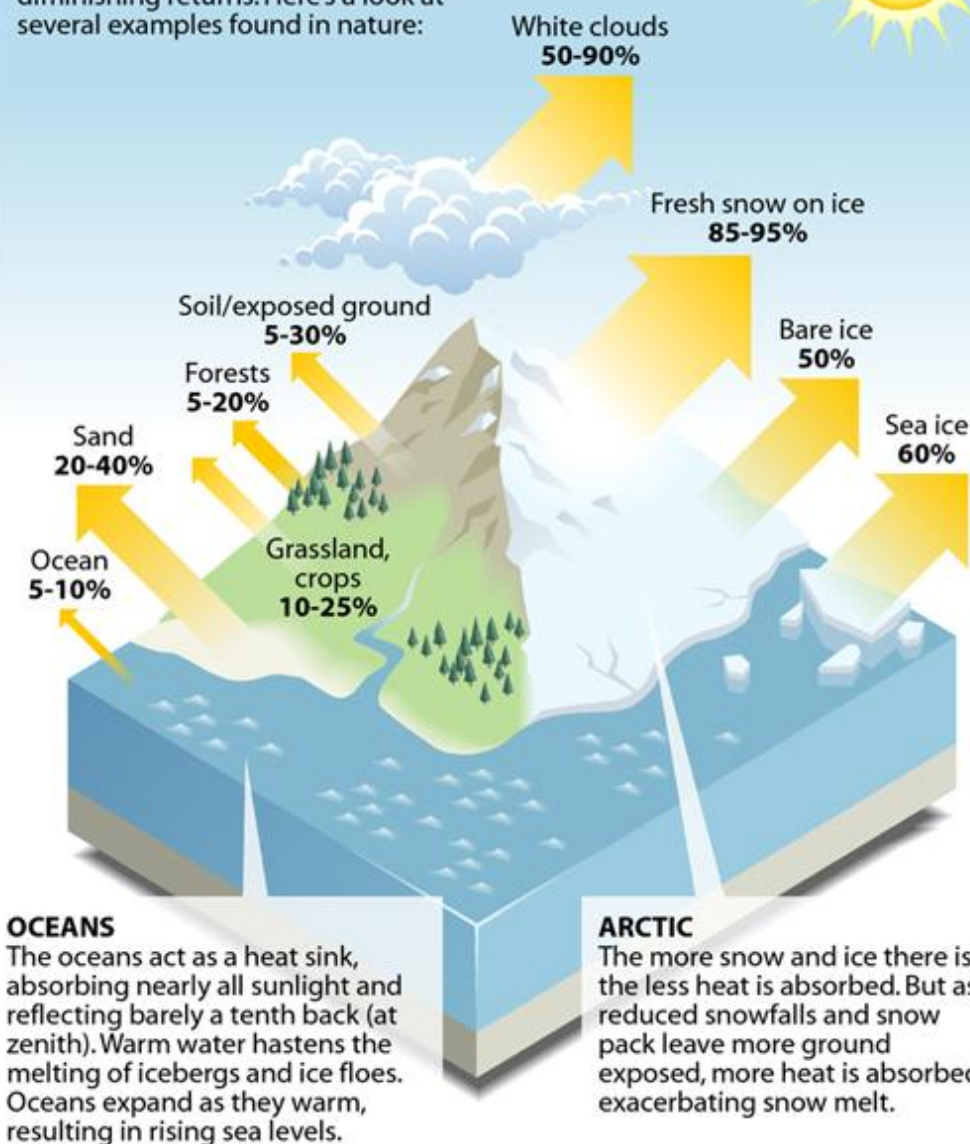
Low Albedo



 reflected
 absorbed
 incoming

Keeping Things Cool: The Albedo Effect

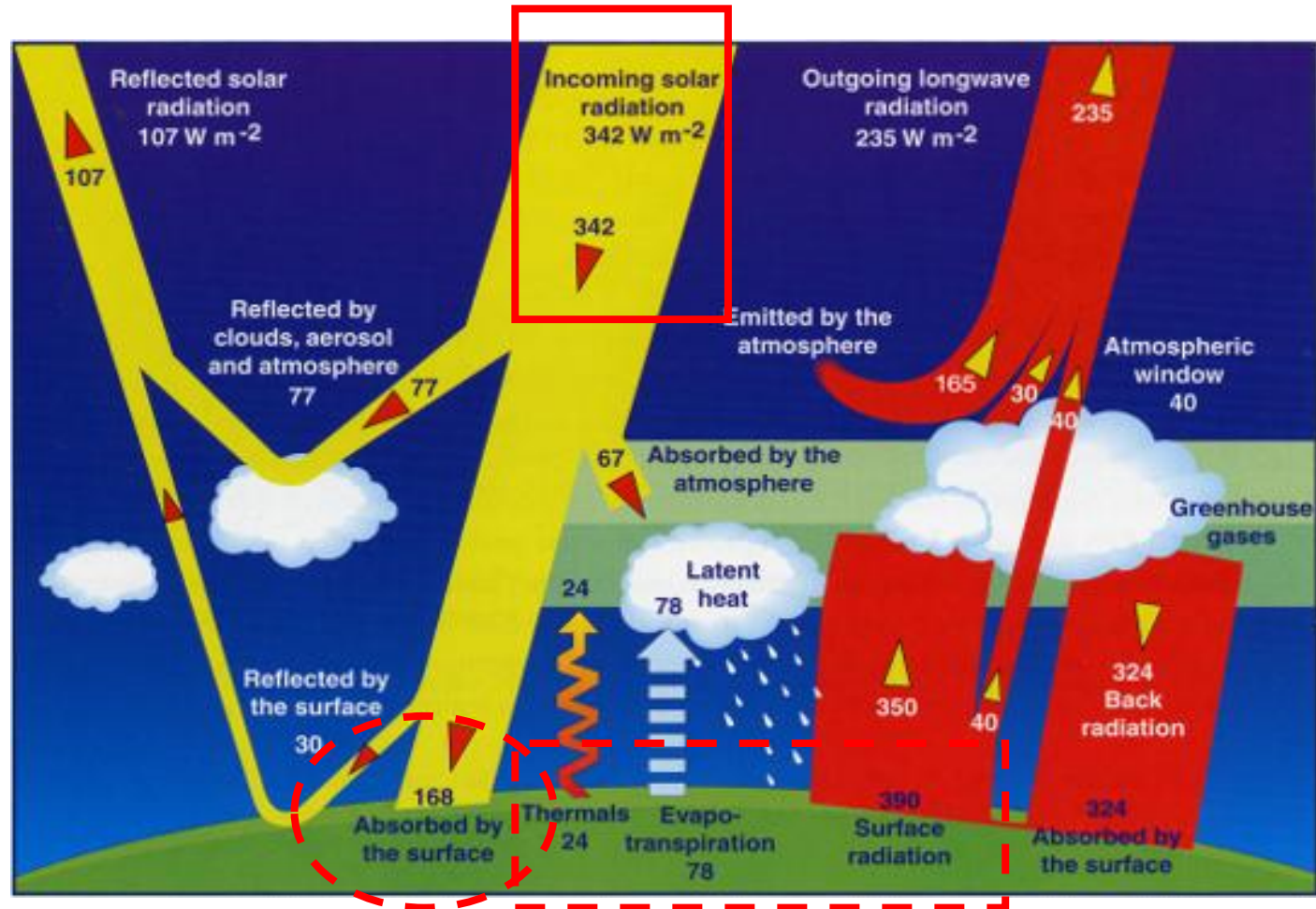
The albedo effect — the reflectivity of sunlight on various surfaces — is important in keeping the Earth cool. Clean, white clouds and fresh snow and ice reflect the most sunlight, while exposed land, water and vegetation have diminishing returns. Here's a look at several examples found in nature:



Note: Illustration is diagrammatic and not to scale.

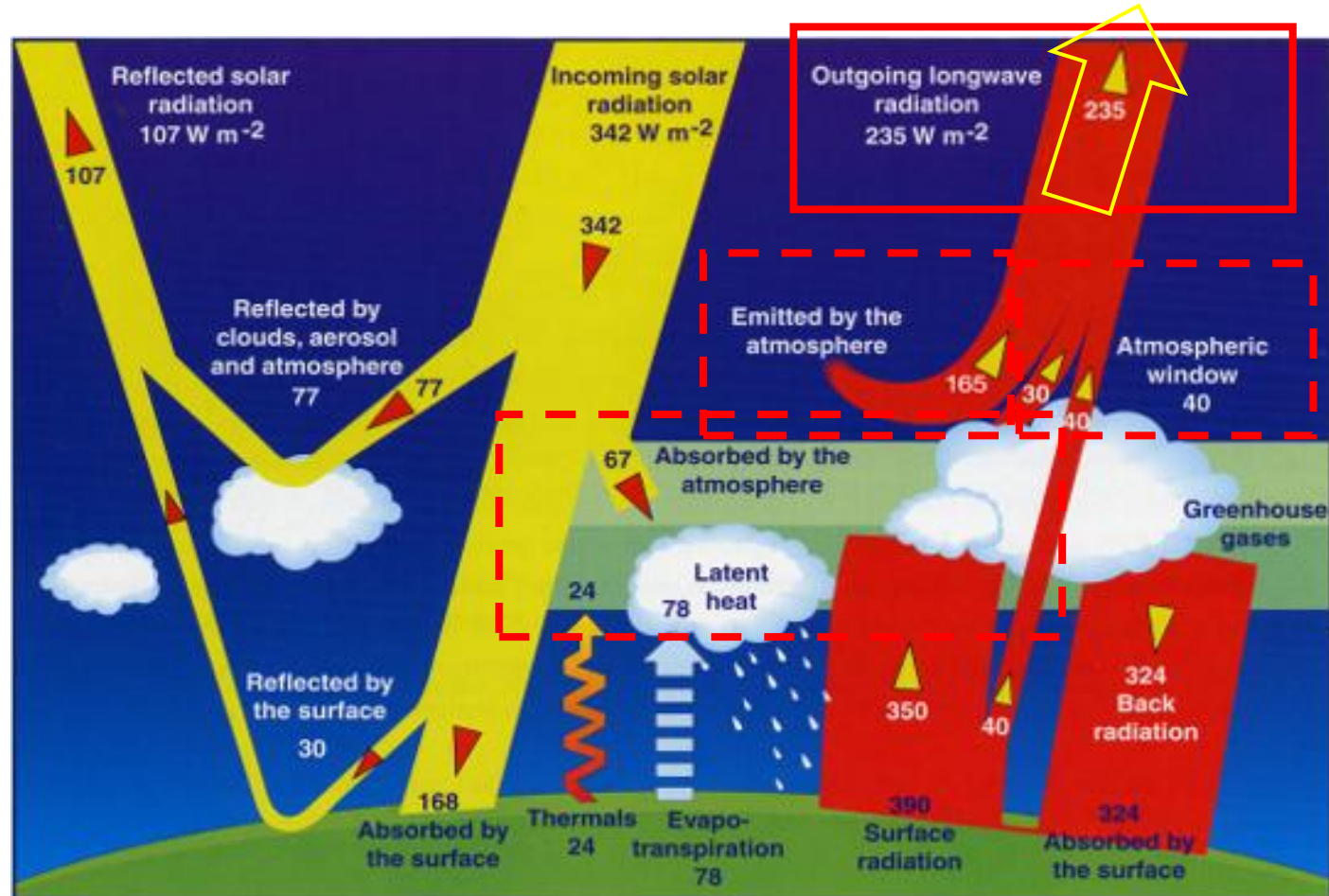
Earth's Energy Balance

When energy comes in, where does it go?



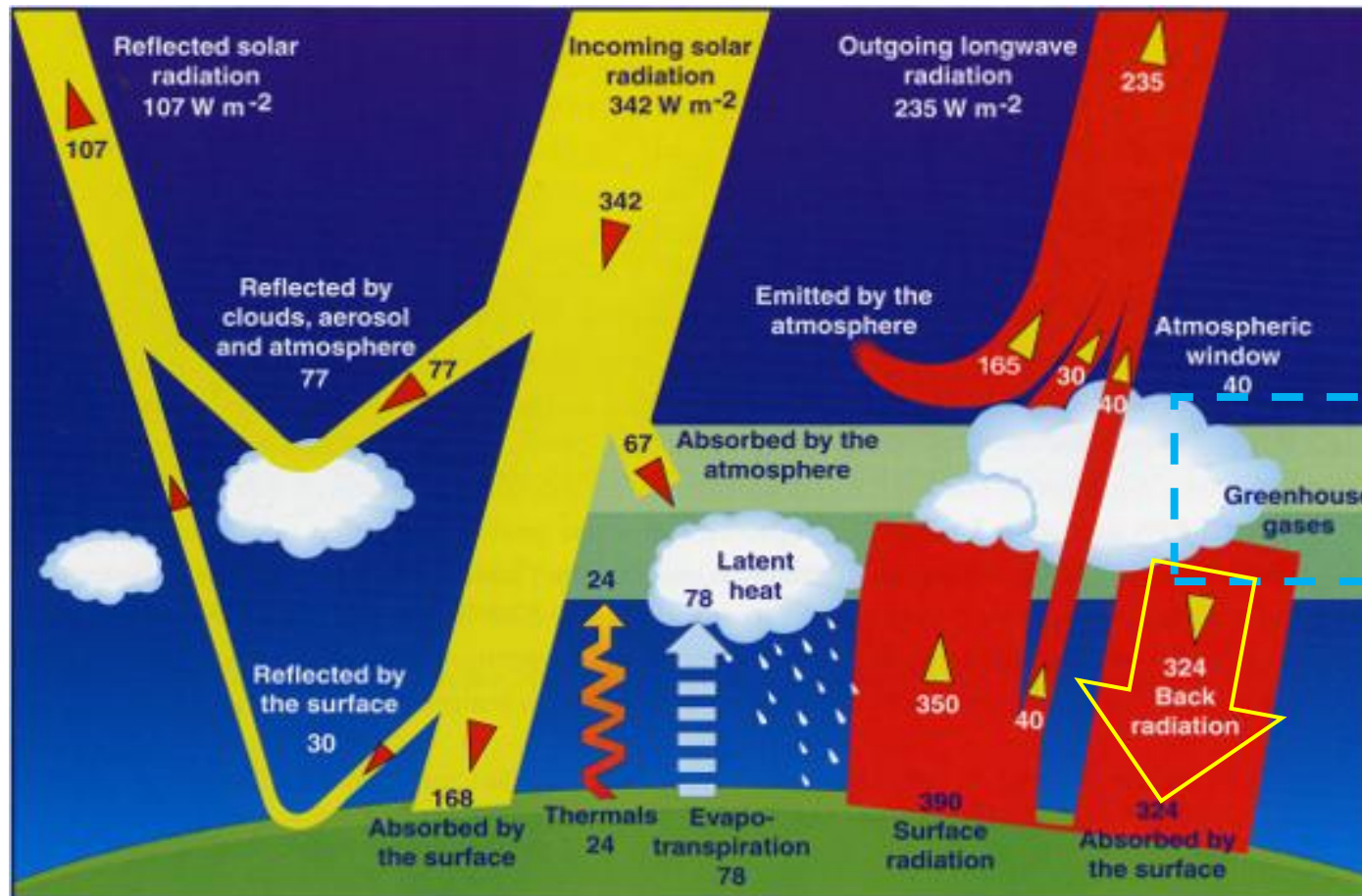
Earth's Energy Balance

When energy is within the atmosphere, where does it go?



Earth's Energy Balance

When energy is within the atmosphere, where does it go?



Greenhouse gases (GHGs)

Make the Earth an **imperfect emitter**

GHG effect:

Process by **which Earth warms** due to the composition of gases in the atmosphere

GHG examples: CO_2 (carbon dioxide), CH_4 (methane), and H_2O_v (water vapor)

Earth's Energy Balance

- Climate (radiative) forcing: a factor that influences Earth's energy balance (natural or anthropogenic)

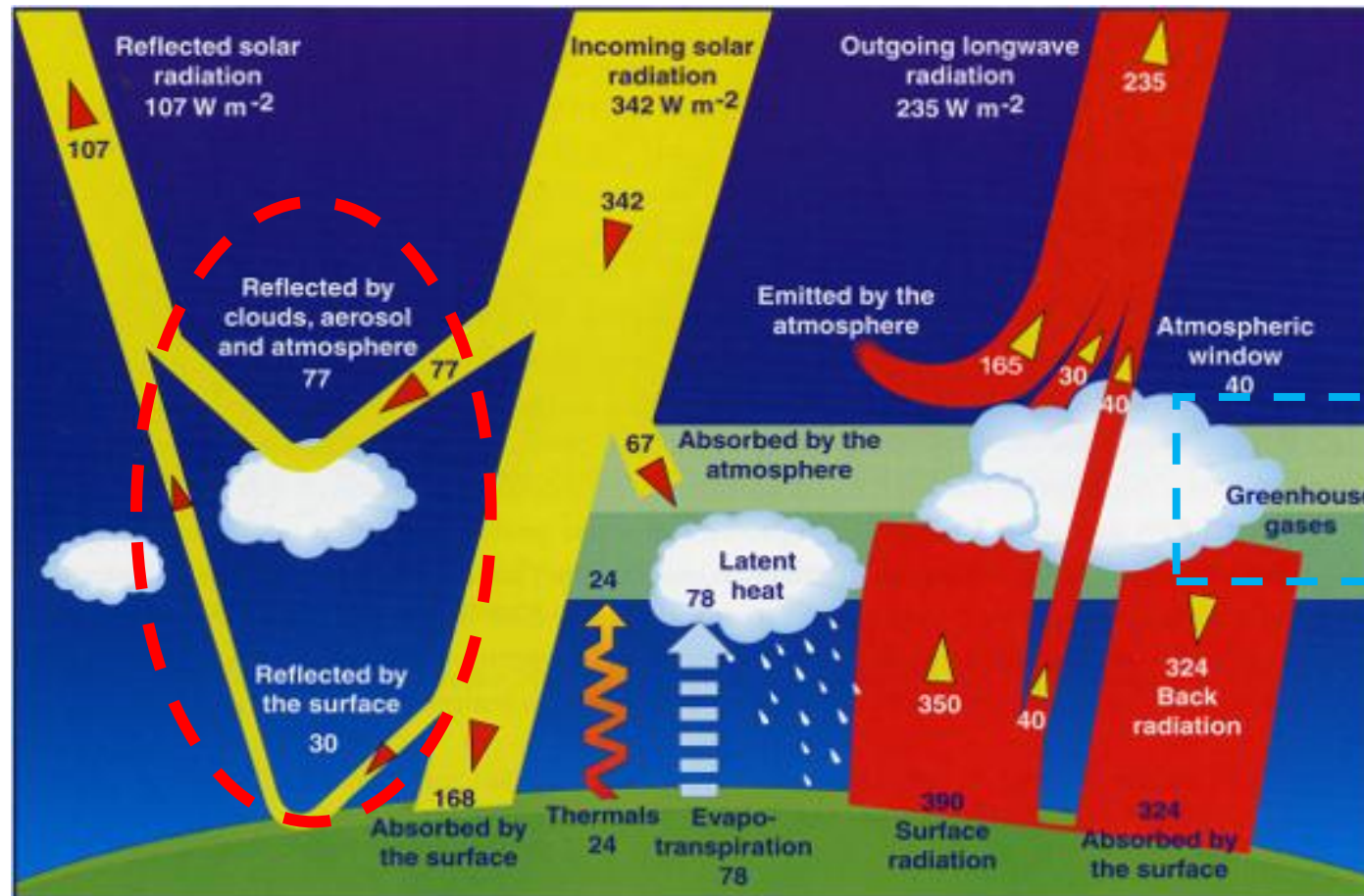
Albedo

Net effect of the **reflectivity** of Earth - makes the Earth **an imperfect absorber**

Reflectivity of Earth: 31% of light is reflected back to space

Albedo examples:

High: ice, snow, clouds
Low: forests, pavements



Greenhouse gases (GHGs)

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GHG effect:

Process by **which Earth warms** due to the composition of gases in the atmosphere

GHG examples: CO_2 (carbon dioxide), CH_4 (methane), and H_2O_v (water vapor)

Energy Balance - Questions

WARM-UP

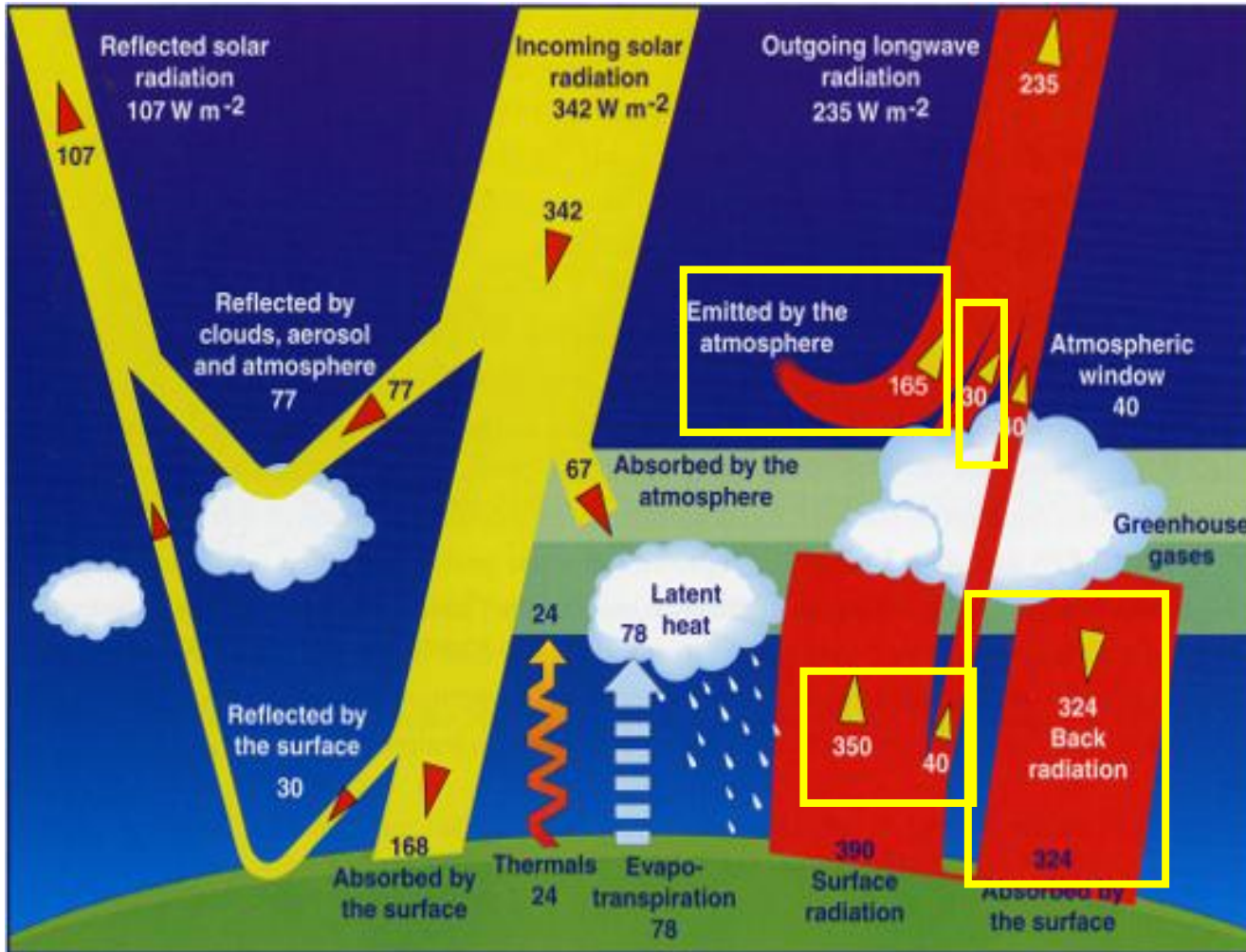
Energy emitted by:
atmosphere? 165 W/m^2
clouds? 30 W/m^2

What is the *surface radiation* (energy emitted by the surface)?

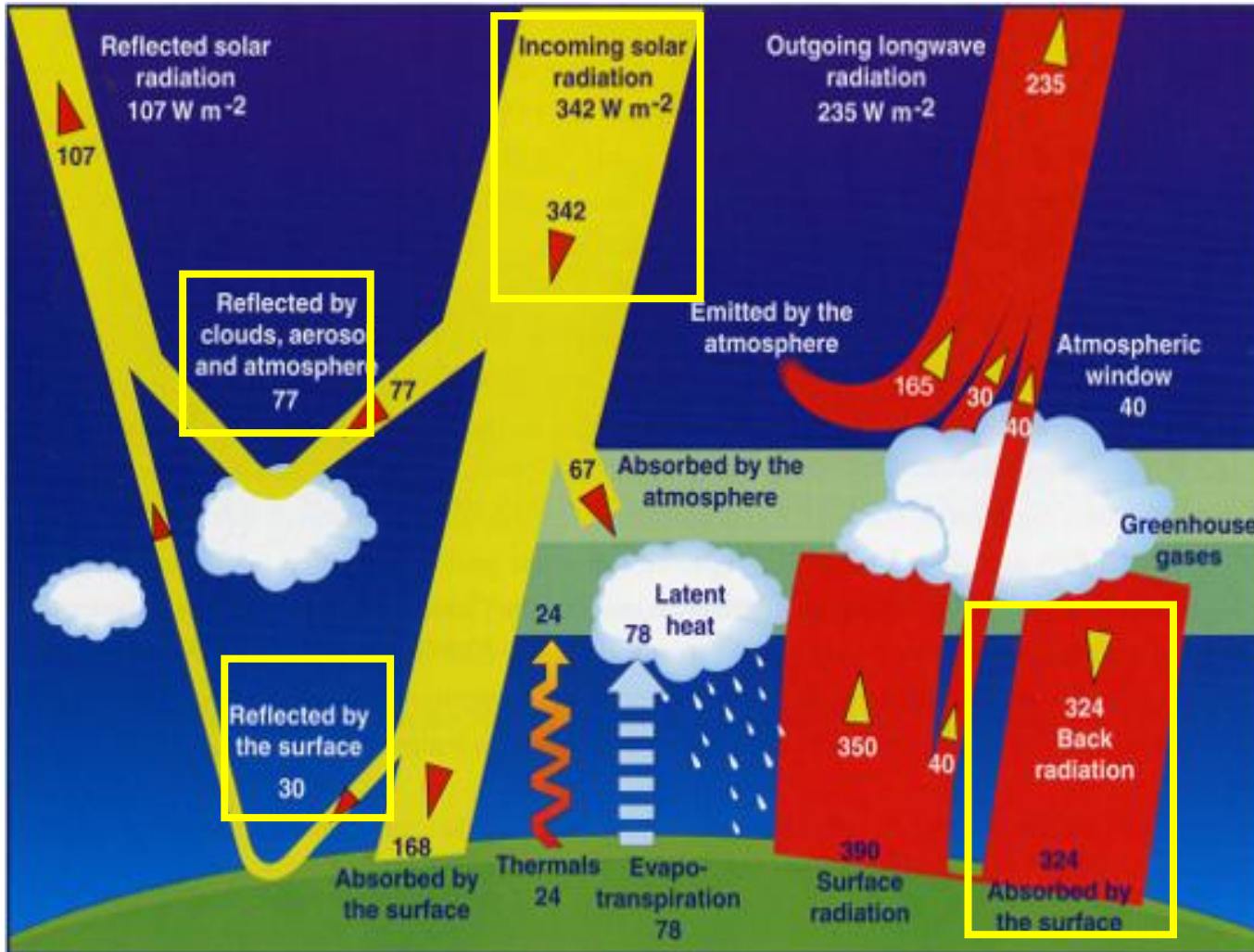
390 W/m^2

How can 390 W/m^2 be emitted by the surface when only 168 W/m^2 is absorbed by the surface from the Sun?

Because there's also 324 W/m^2 in back radiation!
 $168 + 324 = 492 \text{ W/m}^2$



Energy Balance



WARM-UP (cont.)

Calculate how much solar energy is reflected back:

$$30 + 77 = 107 \text{ W/m}^2$$

List different sources of energy absorbed by Earth:

- Incoming solar radiation
- Back radiation from the atmosphere

Which of them has the greatest radiation absorbed by Earth?

Back radiation from the atmosphere

1A. (iii) What values illustrate the balance between energy absorbed at the surface and lost at the surface? Select all that apply and demonstrate the two energy flows are equal with a calculation.

Energy absorbed at the surface:

- a. Solar radiation: 168 W/m^2
- b. Thermals: 24 W/m^2
- c. Surface radiation: 390 W/m^2
- d. Back radiation: 324 W/m^2
- e. Evapotranspiration: 78 W/m^2

Energy lost at the surface:

- a. Solar radiation: 168 W/m^2
- b. Thermals: 24 W/m^2
- c. Surface radiation: 390 W/m^2
- d. Back radiation: 324 W/m^2
- e. Evapotranspiration: 78 W/m^2

Energy absorbed: (a) solar (168) + (d) back radiation (324) = 492 W/m^2

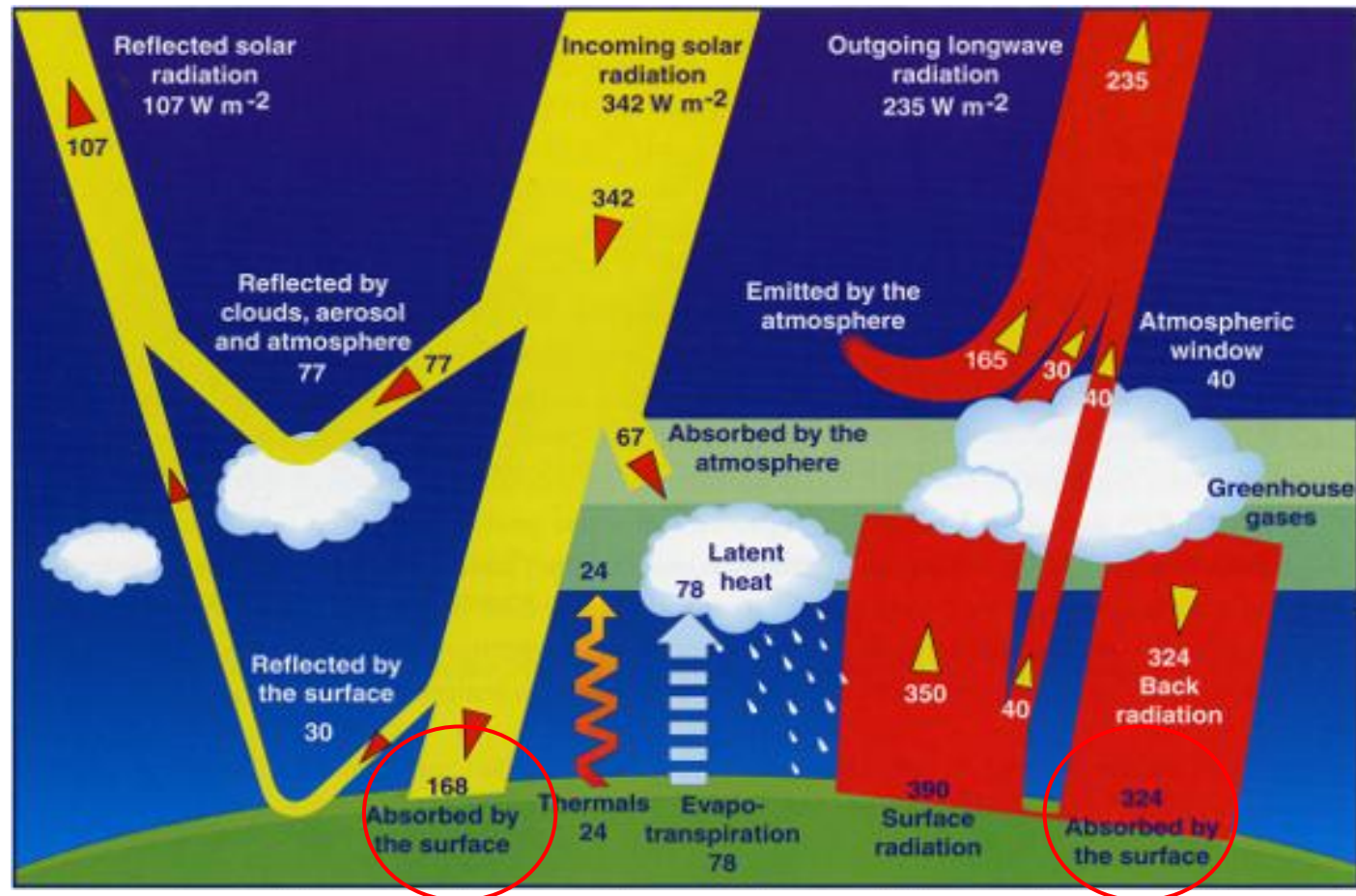
Energy lost: (c) surface radiation (390) + (b) thermals (24) + (e) evapotranspiration (78) = 492 W/m^2

Earth's Energy Balance : Energy Absorbed

Energy In (Shortwave)

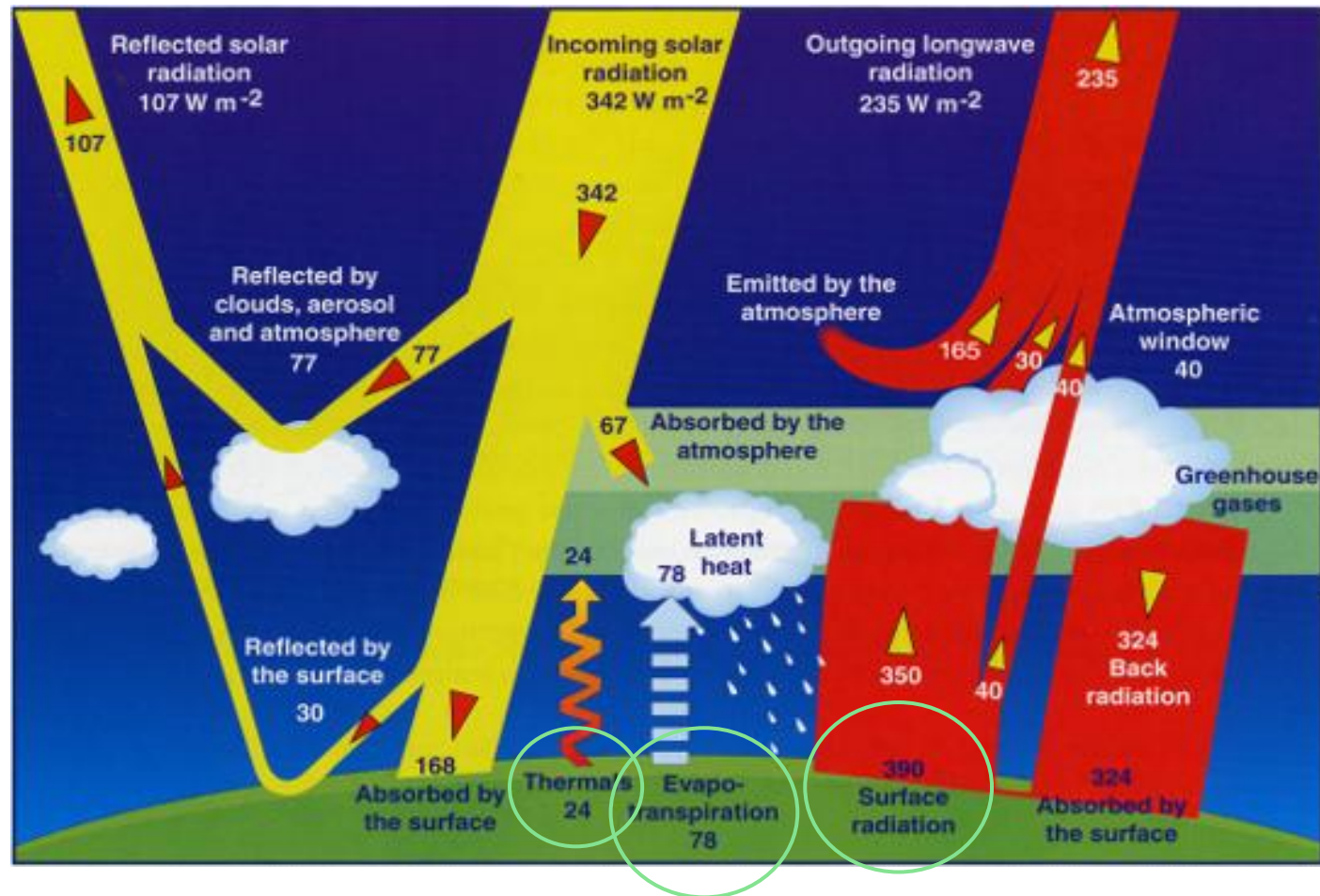
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Energy Out (Longwave)



Earth's Energy Balance : Energy Lost

Energy In (Shortwave) = Energy Out (Longwave)



1B. (ii) Table 1 (below) gives the value of albedo for various surfaces. **Which of the following would cause a decrease in Earth's surface albedo?** Select all that apply.

Substance or Surface	Albedo
Whole Earth average	0.3
Fresh snow	0.8-0.9
White paint	0.5-0.9
Sea ice	0.5-0.7
Desert sand	0.4
Green grass	0.25
Bare soil	0.17
Dark roofs	0.1-0.2
Conifer forest	0.08-0.15
Open ocean	0.06
Fresh asphalt	0.04

***Table 1.** Albedo of various substances and surfaces, defined as the fraction of solar radiation reflected. (Data from Akbari et al., 2012)*

- a. Allowing the size of the Sahara Dessert to increase and take over forests
- b. Melting sea ice
- c. Replacing a grassy field or forested park with a parking lot
- d. Planting grass in open fields of bare soil
- e. Sea level rise increasing the surface area of Earth's oceans
- f. Painting roofs of buildings white

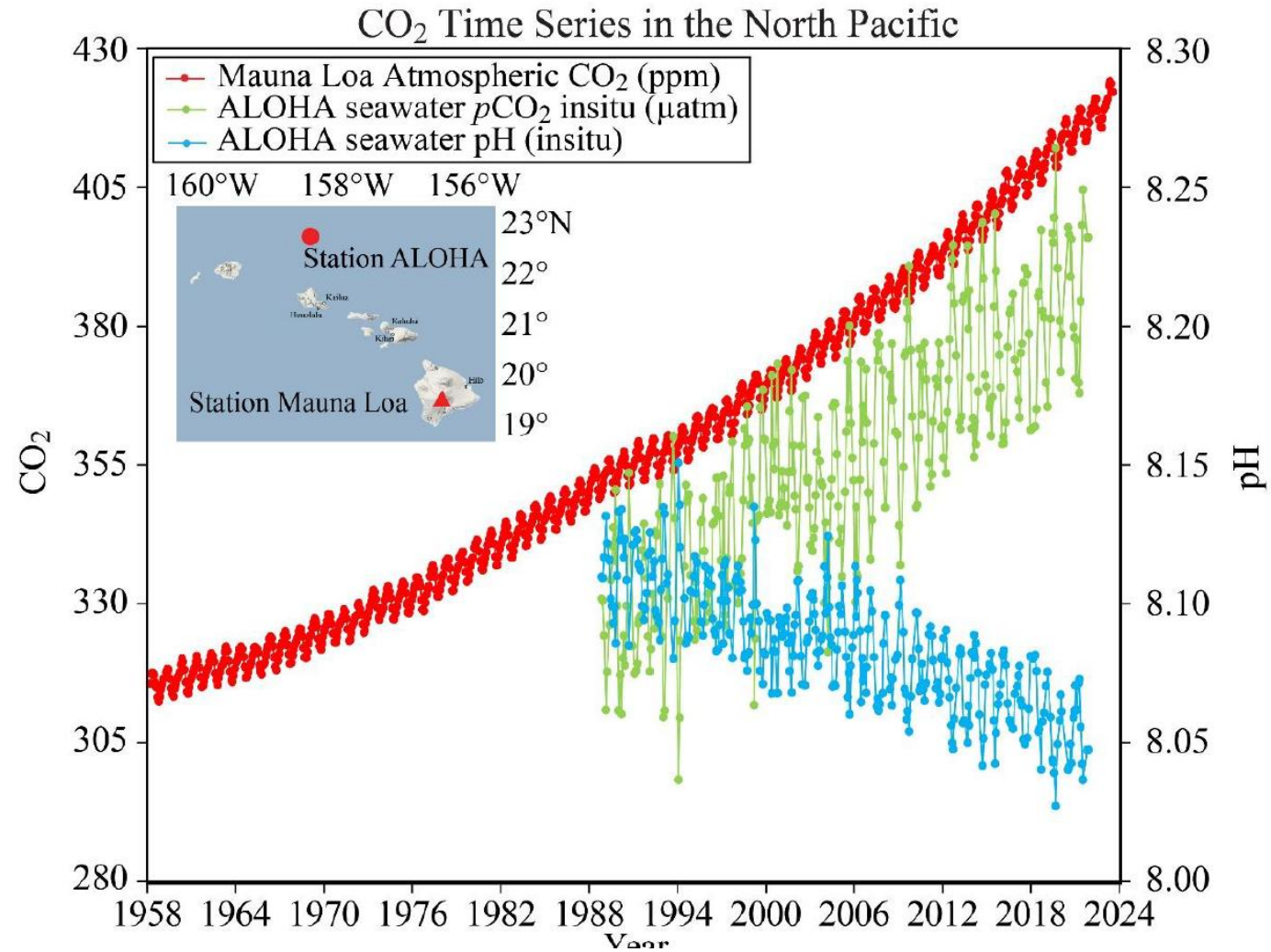
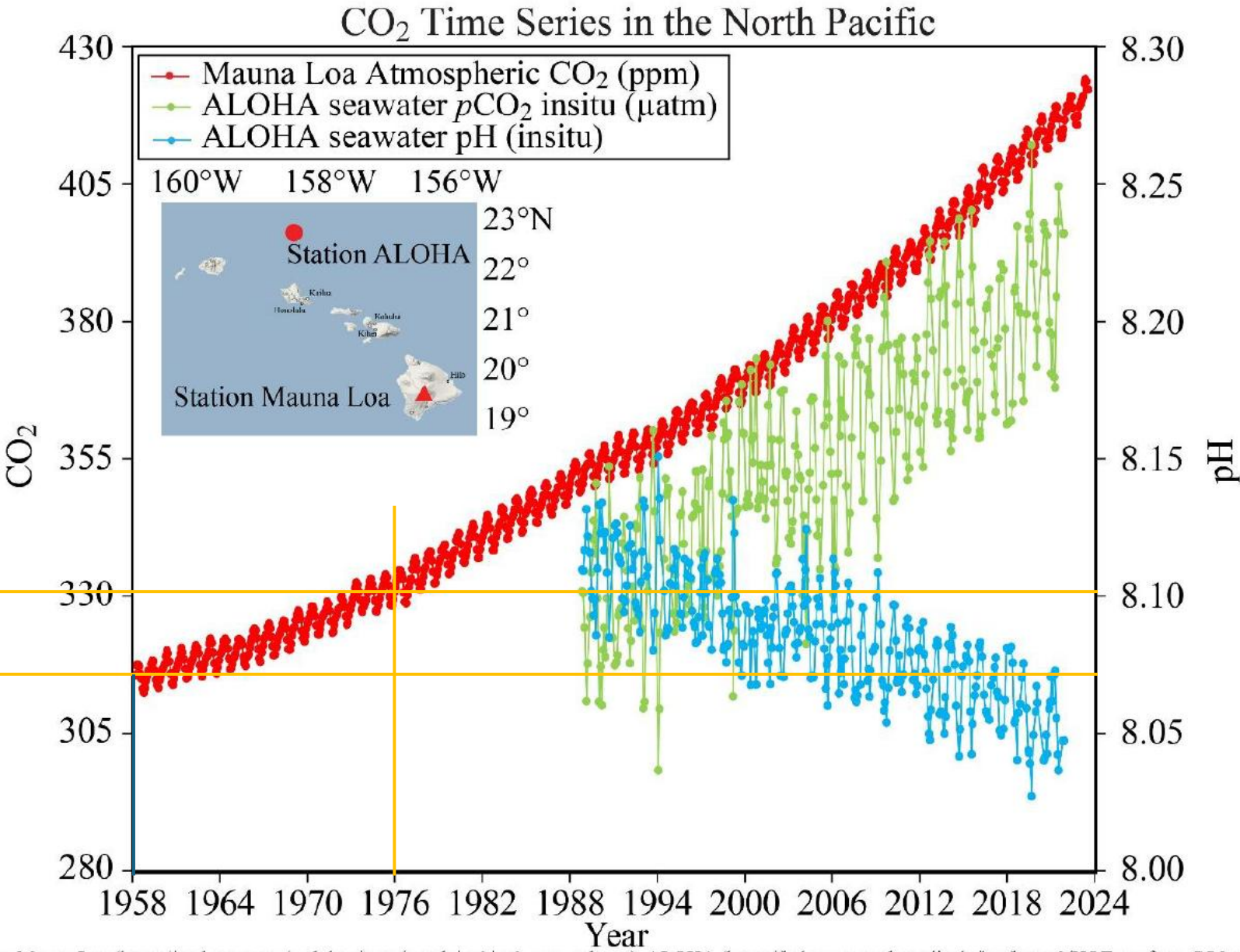


Figure 2. Measurements of atmospheric CO₂ concentration (red line) from the Mauna Loa Observatory station from 1958 to 2024, seawater partial pressure of CO₂ in μatm (green line) and seawater pH (blue line) from station ALOHA from 1990 to February 2024. Source: National Oceanic and Atmospheric Administration Pacific Marine Environmental Laboratory Carbon Program.

2B. (i) Using Figure 2, estimate the annual rate of change for atmospheric CO₂ concentration (in ppm CO₂ per year) at the Mauna Loa Observatory station for the time period from 1958-1976. Show your work, then select the multiple choice answer that is closest to the value you calculated.



- a. 0.05 ppm/year
- b. 0.1 ppm/year
- c. 0.8 ppm/year
- d. 1.6 ppm/year
- e. 2.3 ppm/year

(c) is correct
CO₂ in 1958 = roughly 315 ppm; CO₂ in 1976 = roughly 330 ppm
330 ppm – 315 ppm / 18 years = 0.8 ppm/year.

Isotopes

Isotope = Same element Differing numbers of neutrons

The **isotopic composition of the carbon** in carbon dioxide helps us **identify the source** of the carbon

	C-12 Stable 	C-13 Stable 	C-14 Radioactive 
			
Proton	6	6	6
Neutron	6	7	8
Abundance	98.93%	.07%	trace

$\delta^{13}\text{C}$ notation: used to relate carbon isotope values (C-12 and C-13) on a standardized scale (easy way to compare different sources of CO₂)

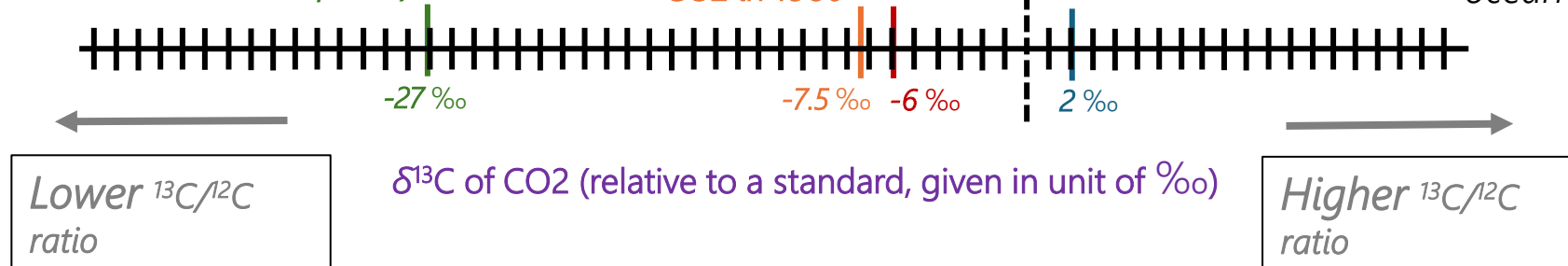
Plants (& plankton) preferentially take up ¹²C because it is lighter

Plants (inc. fossils)

Atmospheric CO₂ in 1980 *Volcanoes*

Dissolved carbon in the ocean

So there is more ¹³C leftover in ocean and atmosphere



2F. In Prof. Helfand's lecture, you learned about the use of isotope ratios to discern the source of atmospheric carbon. Figure 4 below shows the ratio of $^{13}\text{C}/^{12}\text{C}$ in atmospheric CO_2 (relative to a standard), which is denoted as $\delta^{13}\text{C}$. The black line shows the value recorded at the Mauna Loa Observatory, Hawaii, and the red line shows the value recorded at the South Pole, Antarctica.

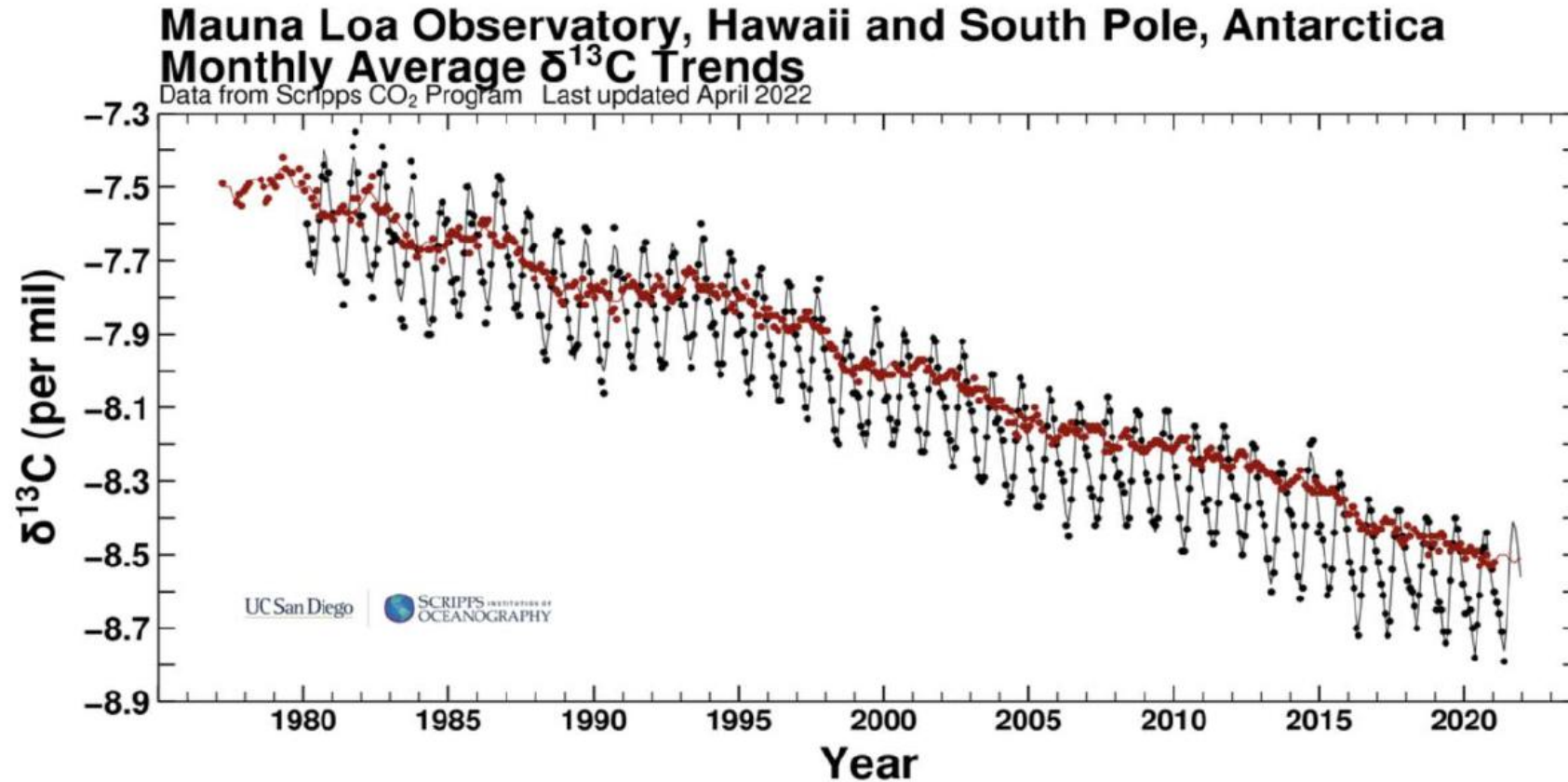
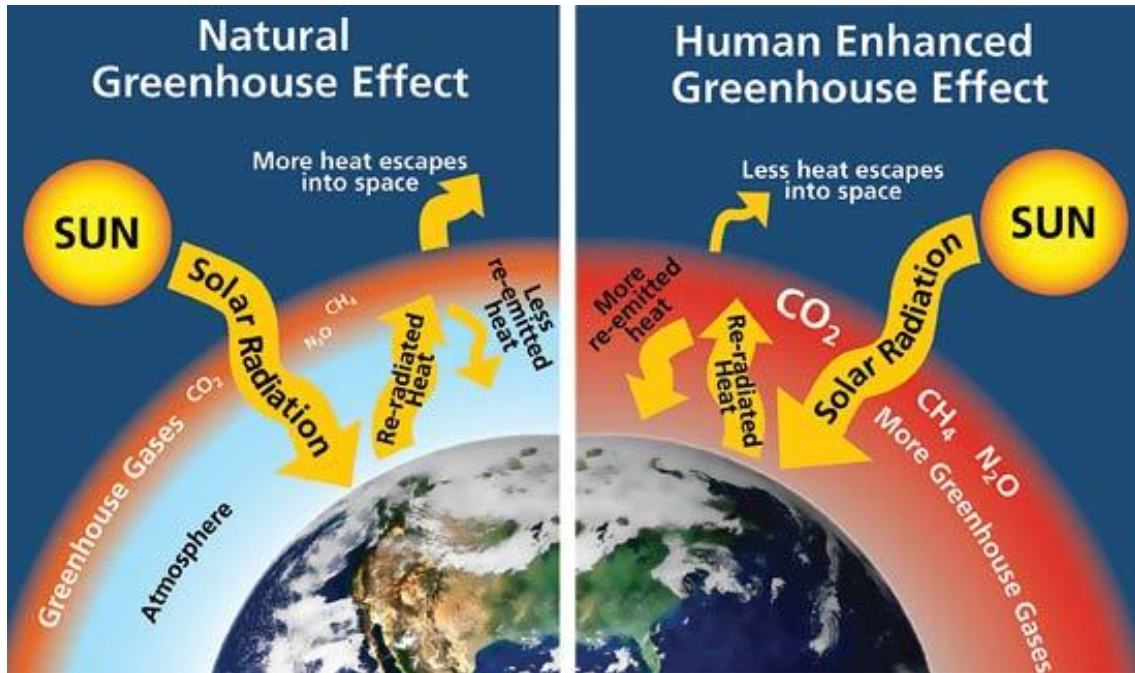


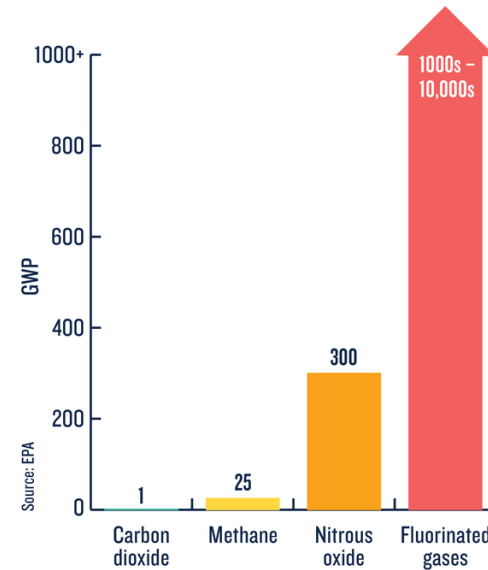
Figure 4. Average monthly $\delta^{13}\text{C}$ records of atmospheric CO_2 from Mauna Loa Observatory, Hawaii (black dots/curve) and South Pole, Antarctica (red dots/curve) from 1975 to 2022. (Note: $\delta^{13}\text{C}$ represents the ratio of $^{13}\text{C}/^{12}\text{C}$ relative to a standard and is recorded in parts per thousand or “per mil”). Source: Scripps CO₂ program.

Earth's Energy Balance, Greenhouse Gasses

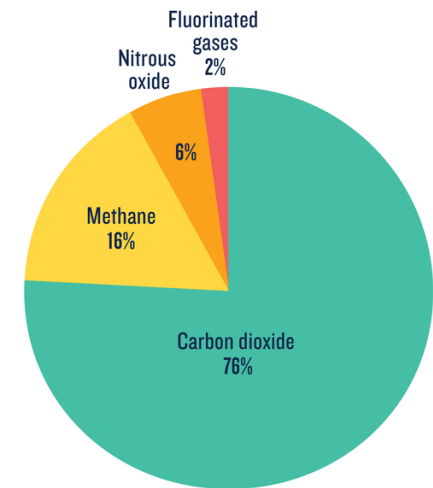
- Other GHG (e.g. N_2O , CH_4) have greater warming potential than CO_2
- But humans release more CO_2



HOW GREENHOUSE GASES WARM OUR PLANET



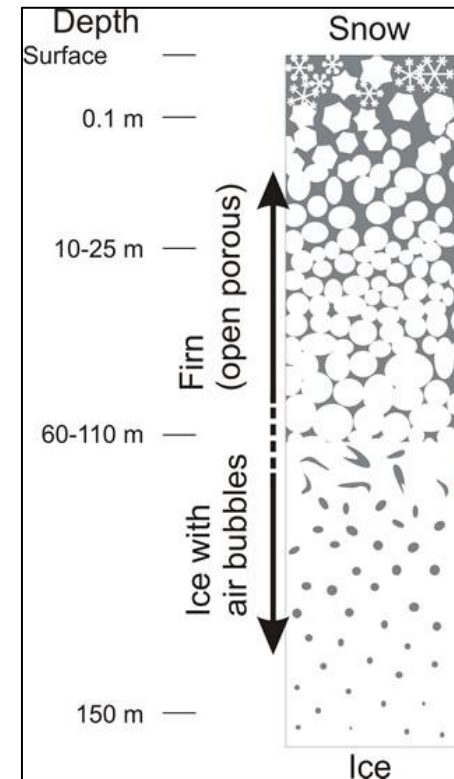
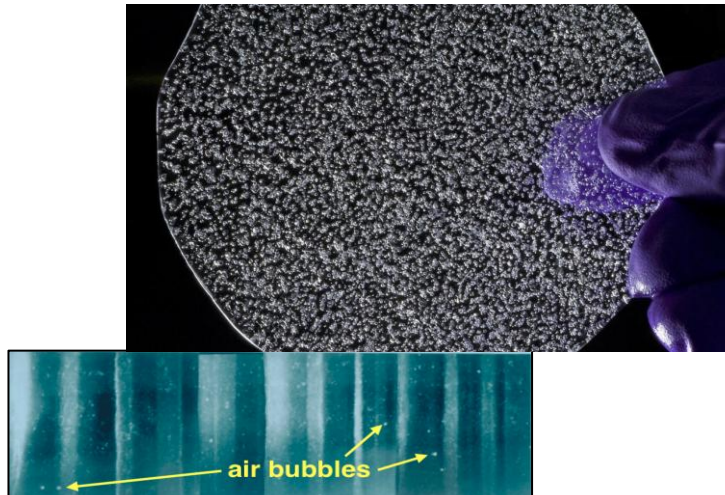
The global warming potential (GWP) of human-generated greenhouse gases is a measure of how much heat each gas traps in the atmosphere, relative to carbon dioxide.



How much each human-caused greenhouse gas contributes to total emissions around the globe.

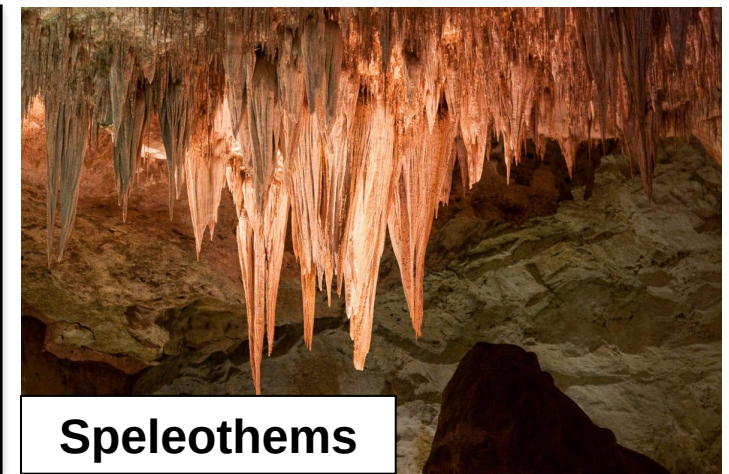
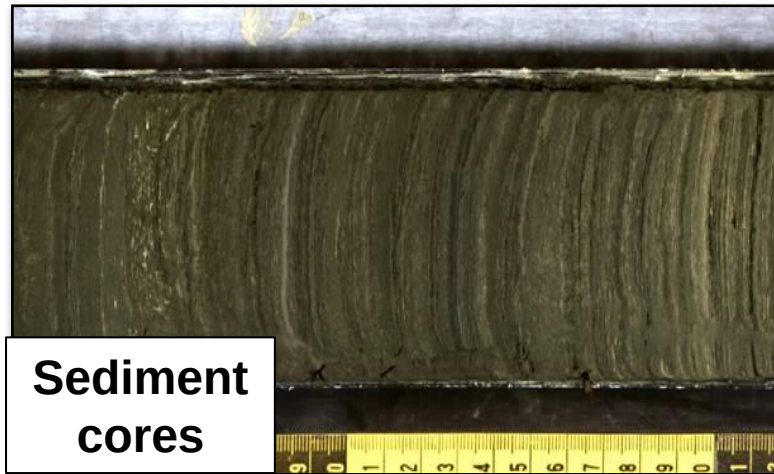
Paleoclimate records: Ice Cores

- Ice cores: trapped air bubbles sample atmosphere each year when it snows
 - Air bubbles can be used to directly determine CO₂ concentration and C isotopes
 - Thickness of annual layers can be used to infer past temperature (qualitative proxy)
 - Oxygen and Hydrogen isotopes are proxies for temperature



Paleoclimate records

- Measurements of air composition in ice cores is one of several methods that reveal past conditions



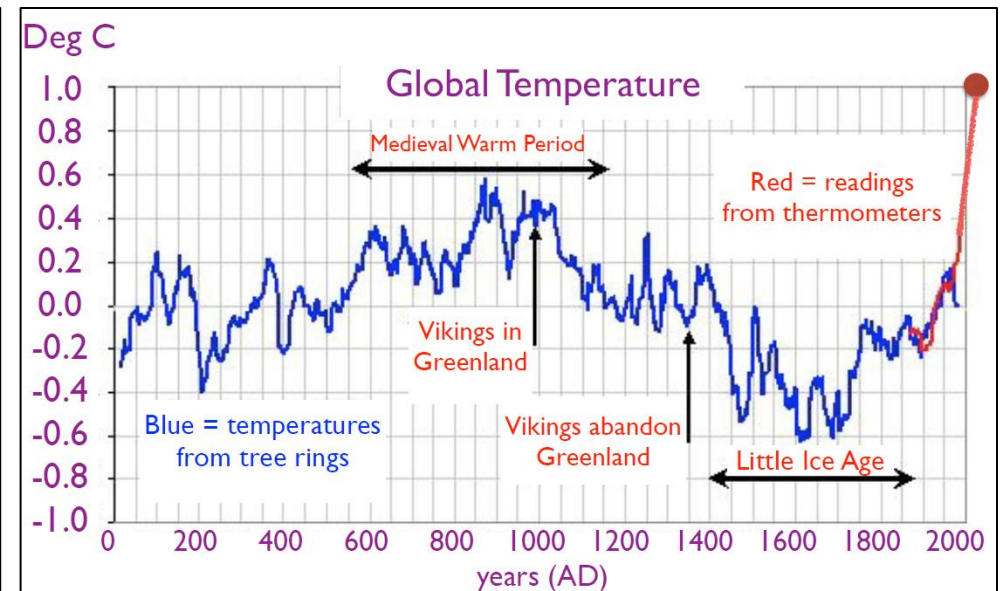
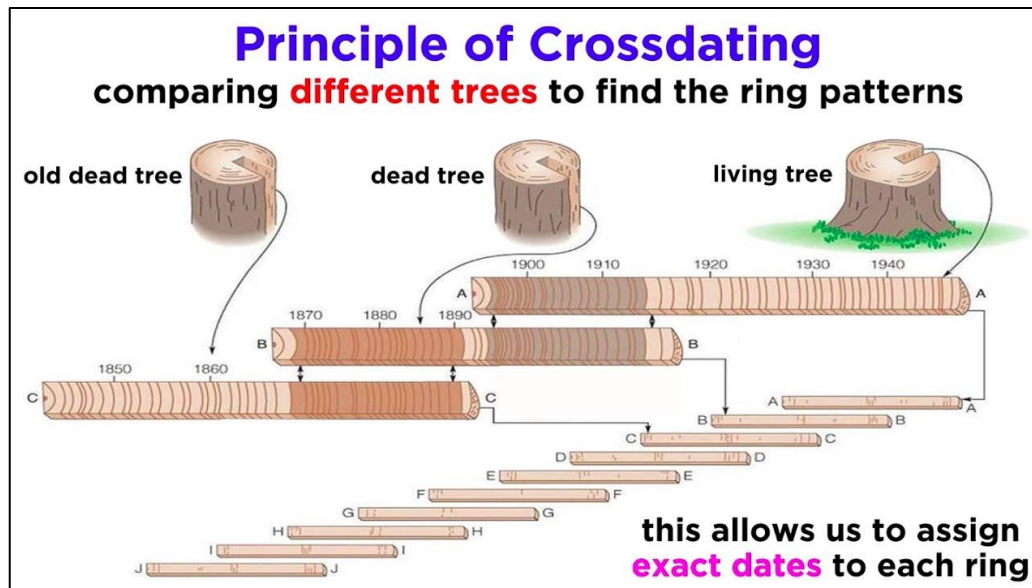
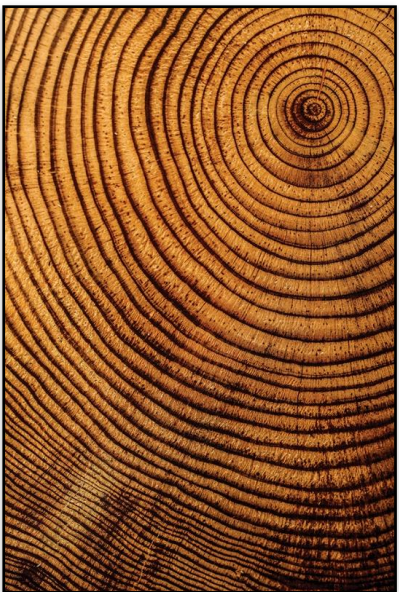
Paleoclimate records

- Different records provide estimates across different time scales

Record	Timescale
Instrumental measurements	150 years
Written historical records	~1500 years
Tree rings	13,910 years
Ice cores	800,000 years
Ocean floor sediments	50 million years
Speleothems	290 million years
Fossils and rocks	4.57 billion years

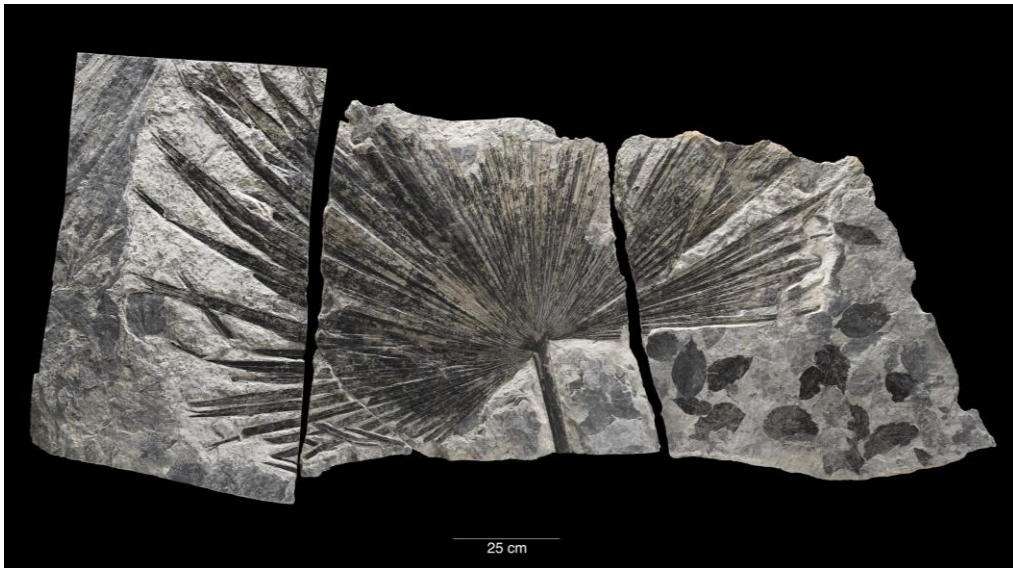
Paleoclimate records: Tree Rings

- Tree rings record conditions each year
 - Oldest trees (Bristlecone pines): 5,000 yrs old, but we can correlate ring pattern & isotope records from different trees with overlap= record for 13,910 yrs
 - Thickness is a proxy for [relative] temperature / precipitation
 - Oxygen isotopes ($^{18}\text{O}/^{16}\text{O}$) is proxy for temperature
 - Carbon isotopes ($^{13}\text{C}/^{12}\text{C}$) is proxy for precipitation

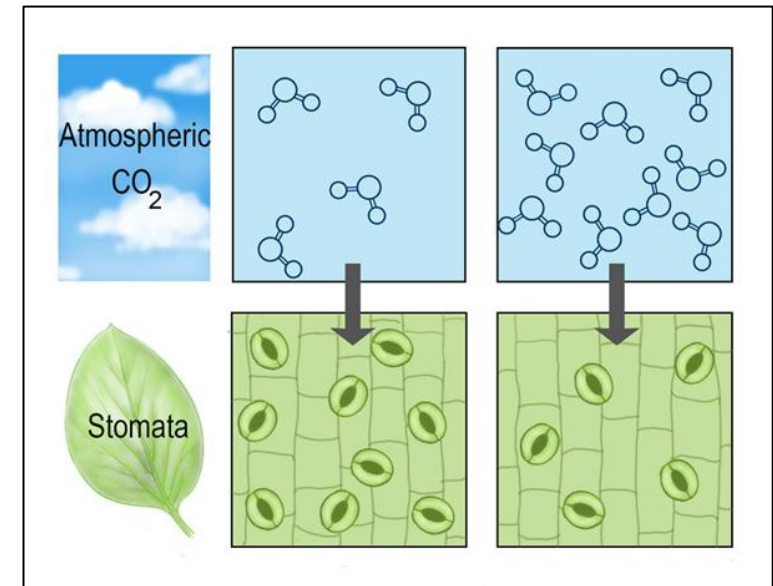


Paleoclimate records: Fossil Plants

- Stomatal density on fossil leaves can be used to infer past CO₂ concentrations
 - Stomata: pores that help the plants take in CO₂
 - lower CO₂ concentrations → harder to get CO₂ → more stomata

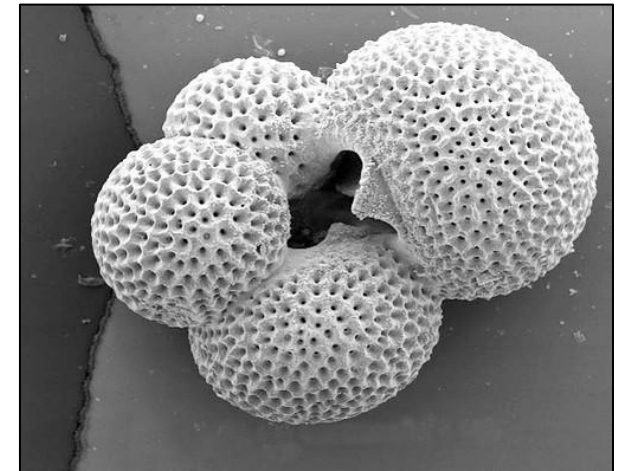
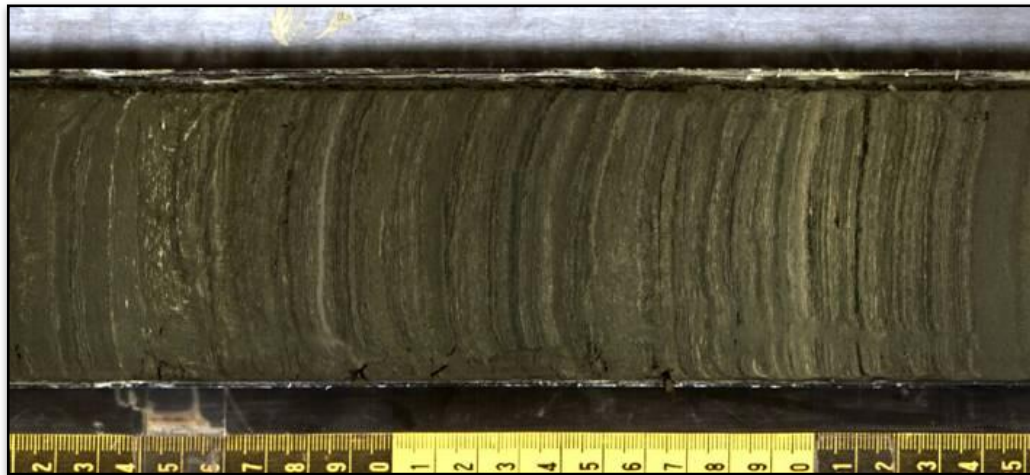


Fossilized leaves from Antarctica- 60 Myrs ago (Smithsonian)



Paleoclimate records: Sediment Cores

- Organic matter deposits in layers on ocean or lake floor (oldest on bottom)
 - O and H Isotope ratios from phytoplankton shells- proxy for temperature
 - Pollen grains can be used to infer species distributions- proxy for climate



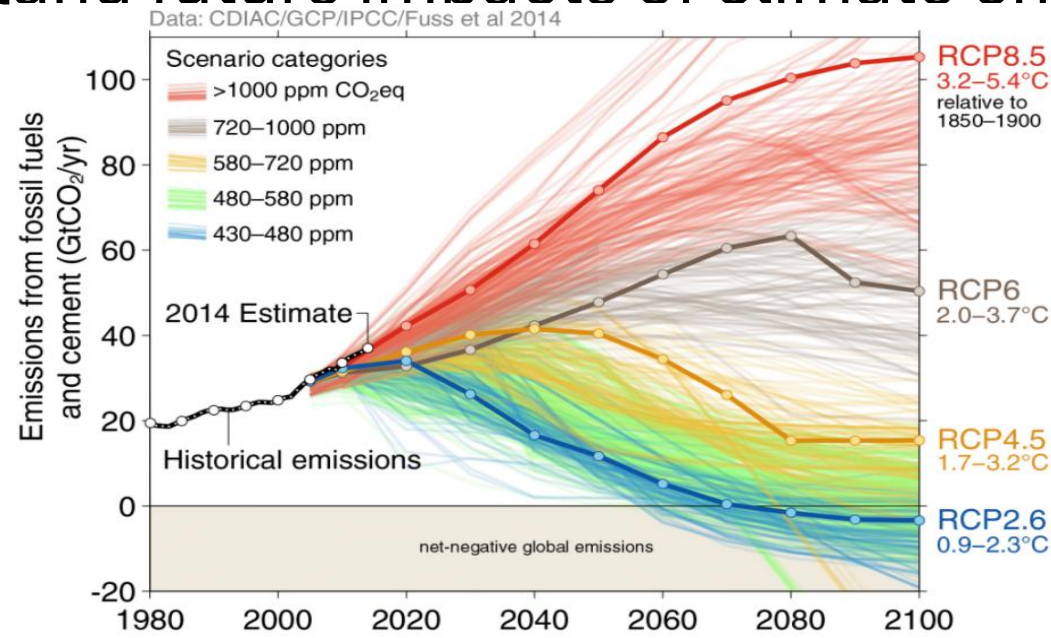
Paleoclimate records: Speliothems

- Layers of CaCO_3 are deposited on stalactites/mites (“speliothems”) in caves



Predicting Future Climate: Modelling

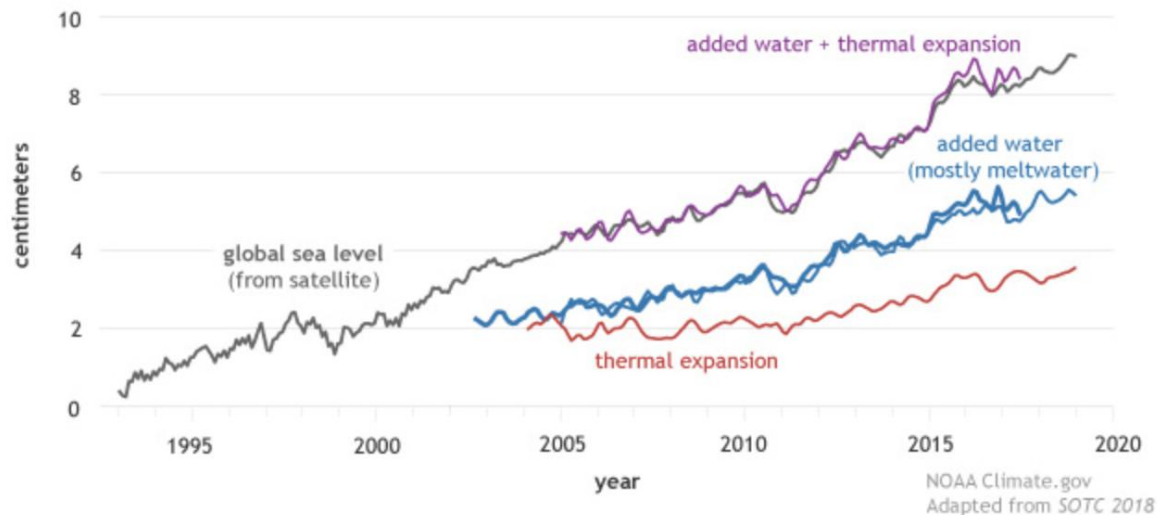
- Climate models generalize processes in the earth system (math. models)
 - to predict how climate will change with further increases in CO₂
 - Validate model by predicting present day/past conditions (hindcasting)
- Used to understand future impacts of climate change, establish climate policy



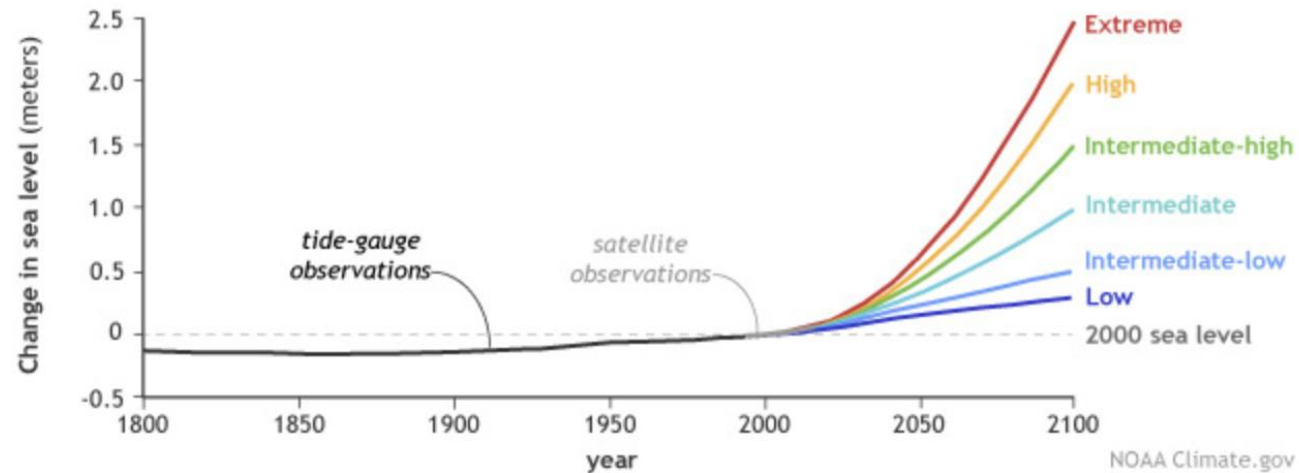
Predicting Future Climate: Modelling

- Modeling Sea Level Rise:
 - Melting glaciers
 - ↑ocean temperature (thermal expansion)

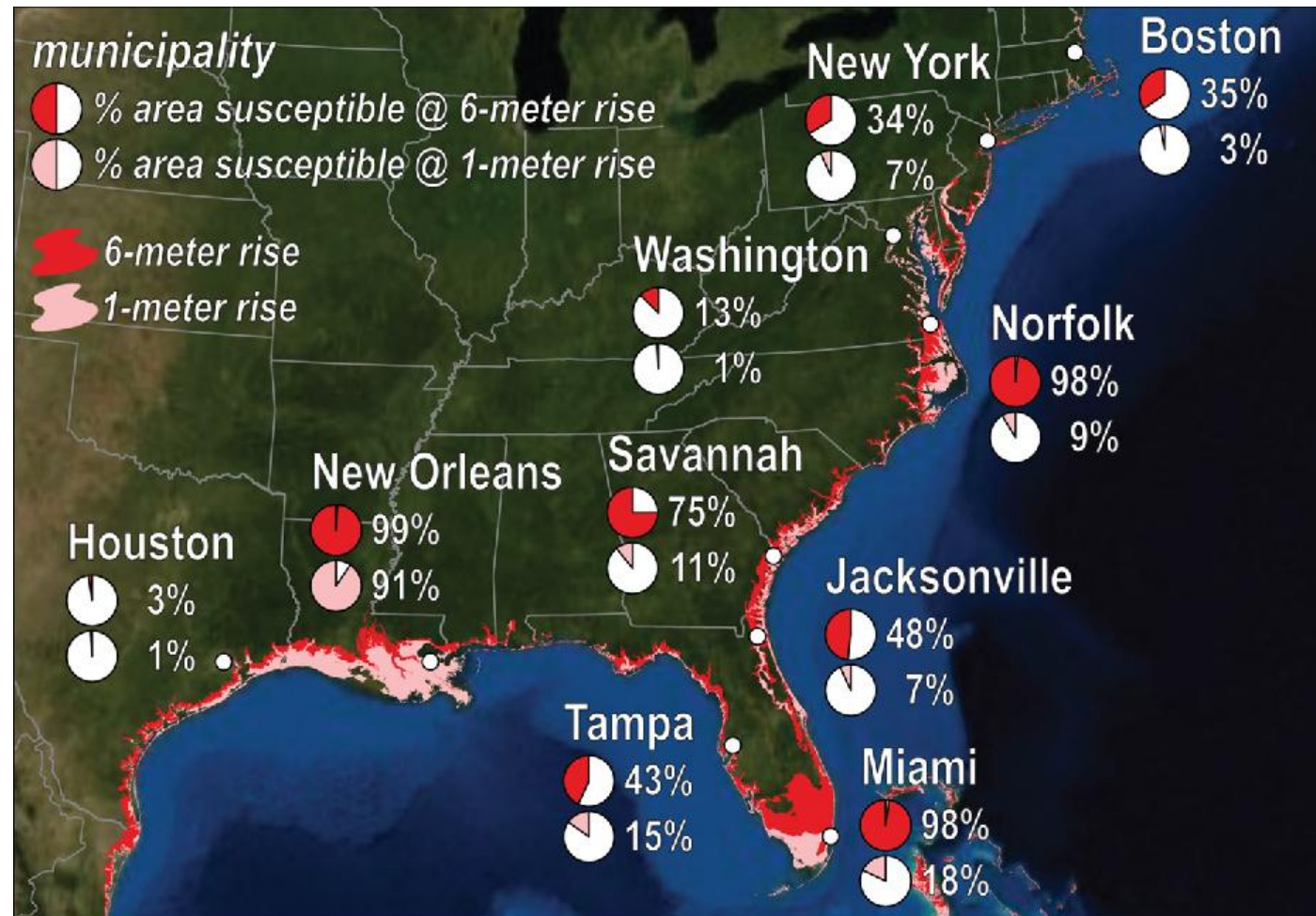
Contributors to global sea level rise (1993-2018)



Possible future sea levels for different greenhouse gas pathways

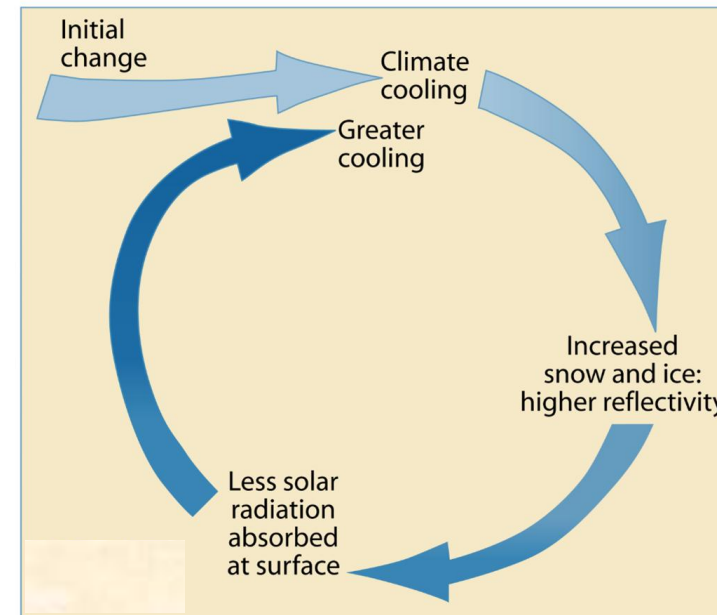
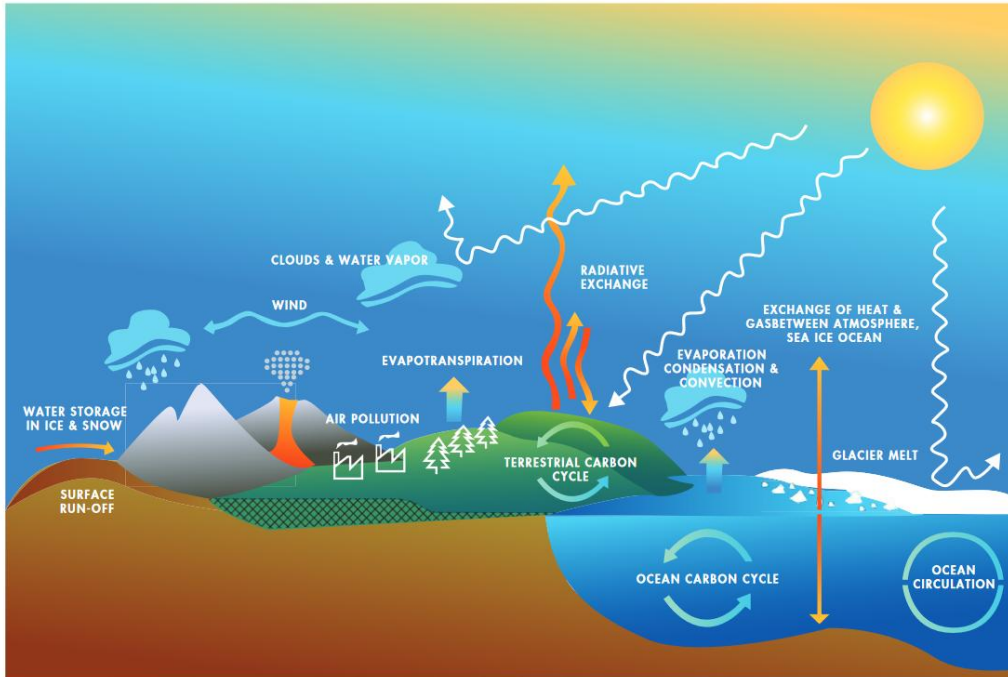


Modelled Effects of Sea-level Rise



Feedbacks

- Feedbacks complicate the models
- Positive: effect amplifies the cause
 - $\uparrow \text{Temperature} = \downarrow \text{Arctic Ice cover} = \downarrow \text{reflected radiation} = \uparrow \text{absorbed} = \uparrow \text{ocean Temp}$
- Negative: effect suppresses the cause
 - $\uparrow \text{Temperature} = \uparrow \text{evaporation} = \uparrow \text{clouds} = \uparrow \text{reflected radiation} = \downarrow \text{Temperature}$



Positive or negative?

Solutions

Mitigation

- Interventions that reduce the sources of greenhouse gases (ex. carbon caps, renewable energy)
- Interventions that enhance the sinks of greenhouse gases (ex. carbon capture)
- Reduce the amount of GHGs in the atmosphere

Adaptation

- Adjustments in response to actual or expected changes in climate
- Aim to reduce harm related to global warming
- Ex. building sea walls to protect from flooding

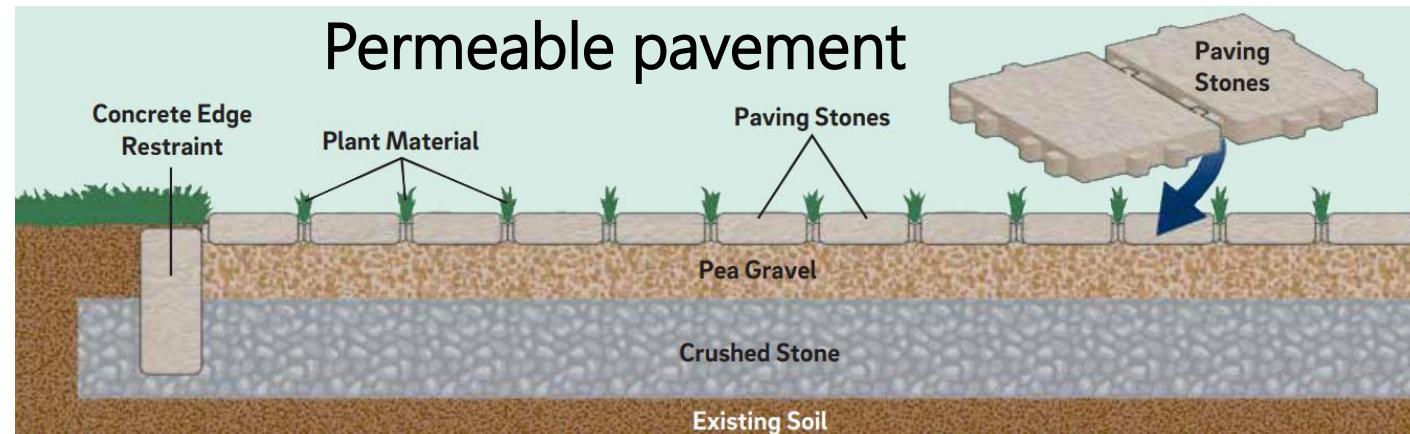
Adaptation Solutions



Bioswale

Green infrastructure bolsters a city's capacity to filter and absorb stormwater

Green roofs



Mitigation Solutions



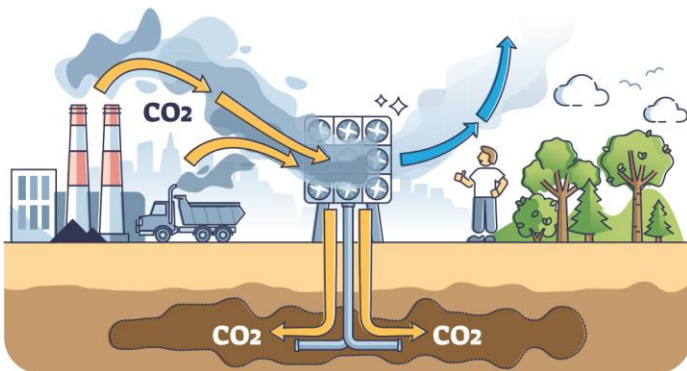
U.S. has invested \$1.2 billion for two pioneering, large-scale carbon capture facilities that will sequester an estimated 1 million tons of CO₂ each per year

Mitigation Solutions

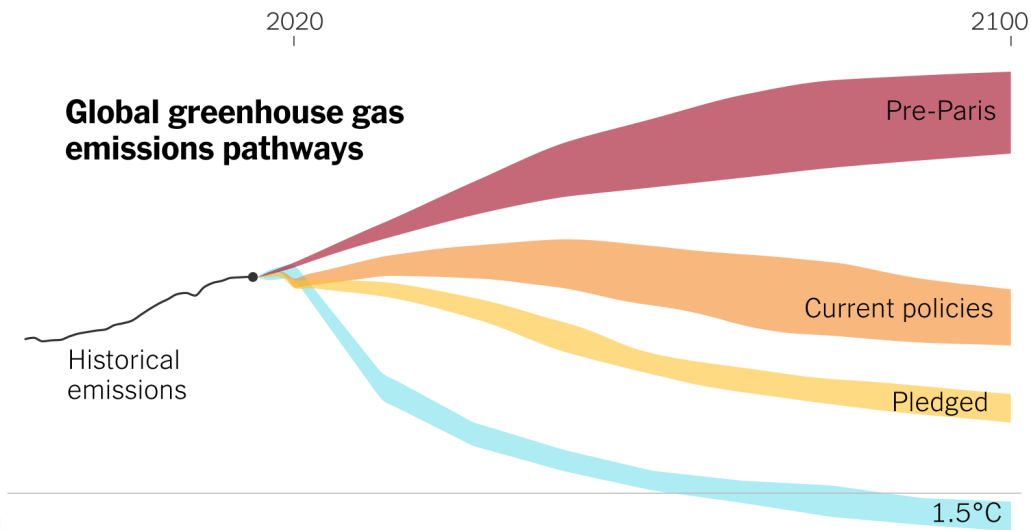
Technological



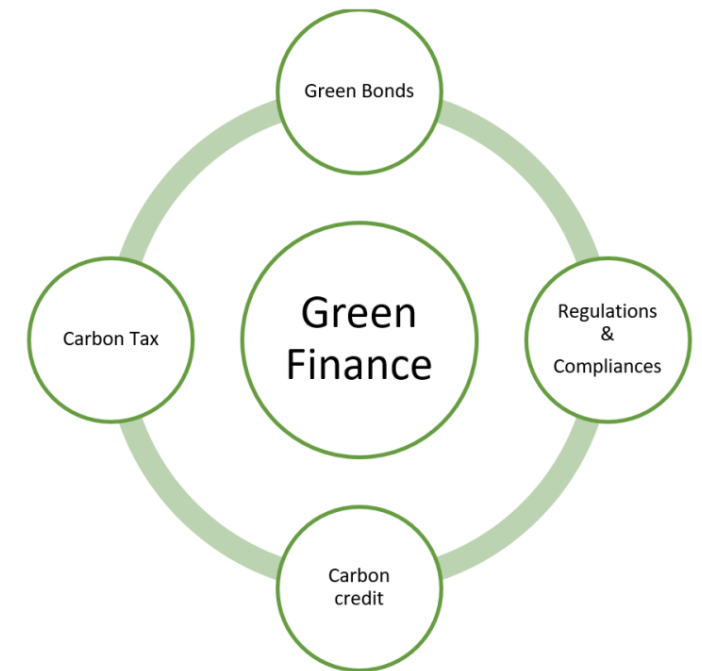
CARBON CAPTURE



Political



Economic



Promise and problems of climate solutions

- Mitigation solutions: technological, political, economic
 - Technological:
 - **Alternative energy**: solar, wind, hydrothermal, geothermal, and nuclear power
 - **Electrification/ increased efficiency** of transportation, buildings and industry
 - **Carbon capture** (requires a lot of energy currently)
 - Changing Earth's energy balance via geoengineering (sulfate aerosols= more clouds)- consequences??
 - Political:
 - Creating and adhering to or enforcing global agreements (e.g. 2015 Paris Agreement)
 - “reducing temperature rise to <2°C and adapt to new conditions that face our planet”
 - **Land use: Reduce deforestation; reduce agricultural emissions** of methane and NO₂ (fertilizer)
 - Economic:
 - Green Bonds: for development of underutilized/abandoned buildings into qualified green buildings
 - **Carbon Taxes**
 - Green technology becomes more affordable (E-Vs: 26% drop in price every year since 2010) as it becomes more popular and production increases

*Green = parameters
in modelling activity